

(12) **United States Patent**
Chou et al.

(10) **Patent No.:** **US 12,397,517 B2**
(45) **Date of Patent:** **Aug. 26, 2025**

(54) **VEHICLE-MOUNTED AIR COMPRESSOR**

(56) **References Cited**

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U.S. PATENT DOCUMENTS

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8,978,717 B2 * 3/2015 Nakao B29C 73/166
141/38
11,034,104 B2 * 6/2021 Dowel B29C 73/166
2003/0024596 A1 * 2/2003 Kojima B65D 47/36
141/38
2003/0056851 A1 * 3/2003 Eriksen B29C 73/166
141/105
2013/0092286 A1 4/2013 Chou

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(Continued)

FOREIGN PATENT DOCUMENTS

(21) Appl. No.: **18/614,766**

CN 102791470 11/2012
CN 209569133 11/2019
CN 209582360 11/2019

(22) Filed: **Mar. 25, 2024**

(Continued)

(65) **Prior Publication Data**

US 2025/0128679 A1 Apr. 24, 2025

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Related U.S. Application Data

(60) Provisional application No. 63/592,553, filed on Oct. 23, 2023.

(30) **Foreign Application Priority Data**

Feb. 22, 2024 (TW) 113106360

(51) **Int. Cl.**

B29C 73/16 (2006.01)

B60S 5/04 (2006.01)

(52) **U.S. Cl.**

CPC **B29C 73/166** (2013.01); **B60S 5/04** (2013.01)

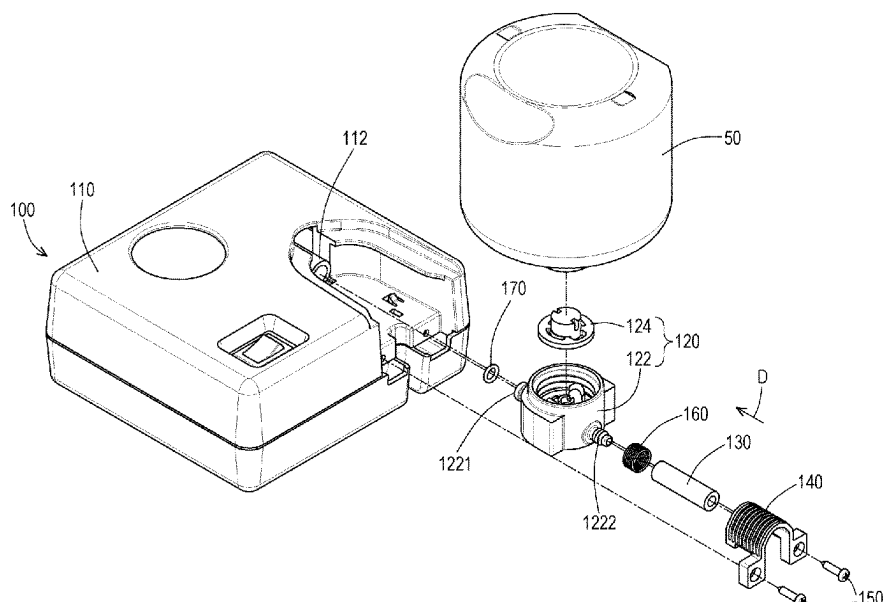
(58) **Field of Classification Search**

CPC B29C 73/166; B65D 83/40
See application file for complete search history.

(57) **ABSTRACT**

A vehicle-mounted air compressor includes an air compressor main body and a bottle cap assembly including bottle cap and sealing and flow guiding components. The bottle cap component is at the air compressor main body, has air inlet/outlet openings, and connected to the air compressor main body via the air inlet opening. The sealing and flow guiding component is at the bottle cap component and includes sealing and flow guiding portions, the former surrounding the latter, and the latter covering the air inlet opening and having at least one air inlet hole and an air outlet hole and connected to the air inlet opening via the air inlet hole and connected to the air outlet opening via the air outlet hole. An airflow from the air compressor main body passes through the air inlet opening, the air inlet hole, the air outlet hole, and the air outlet opening in sequence.

18 Claims, 23 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2018/0272631 A1 9/2018 Zaum et al.

FOREIGN PATENT DOCUMENTS

DE	102013114853	6/2015
DE	102019217775	5/2021
JP	2010184354	8/2010
JP	2011131547	7/2011
JP	2012188109	10/2012
JP	2017056662	3/2017
JP	2023526050	6/2023
JP	3246628	5/2024
JP	3247493	7/2024
TW	I813367	8/2023
TW	M657520	7/2024
WO	2013047419	4/2013

* cited by examiner

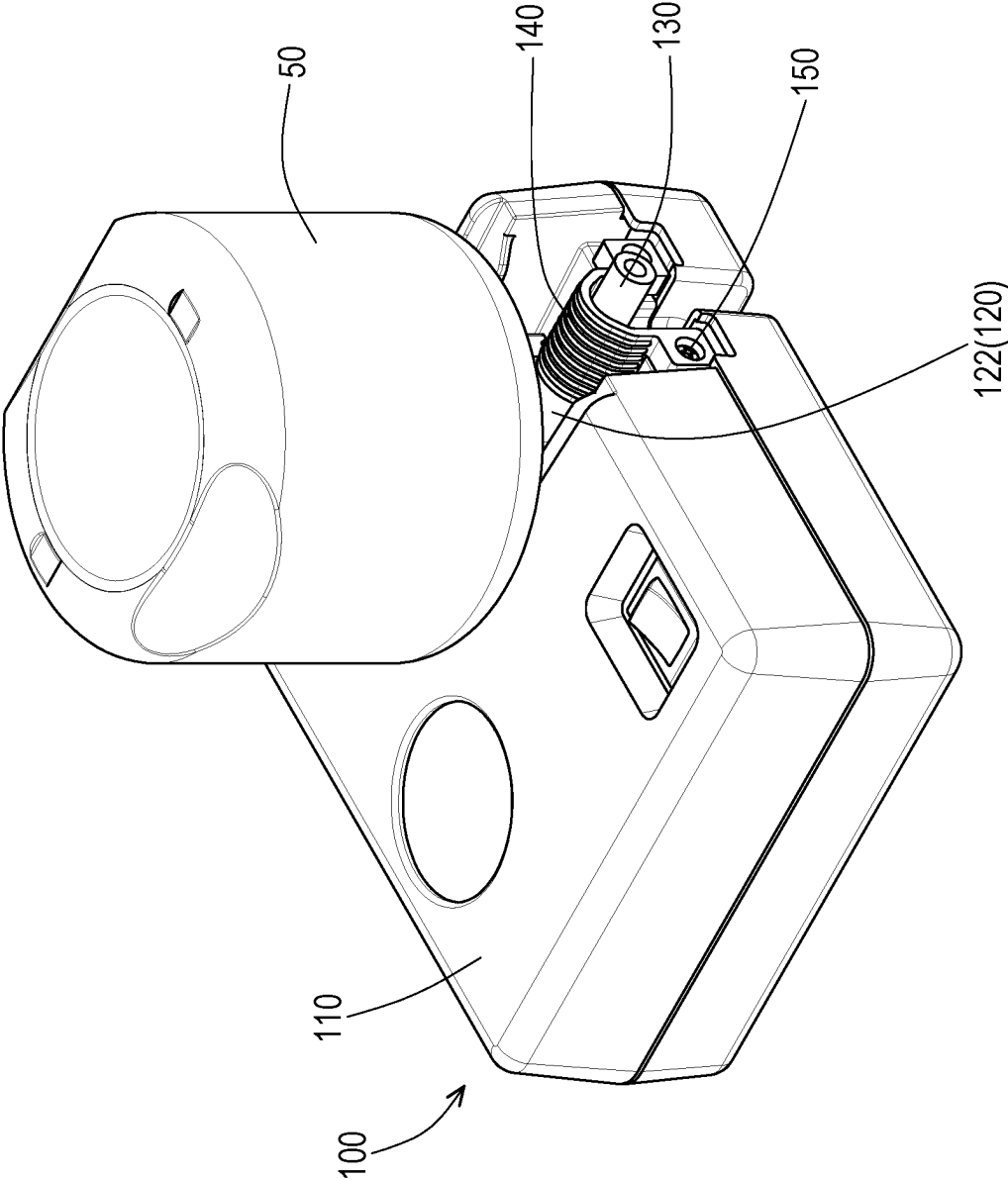


FIG. 1

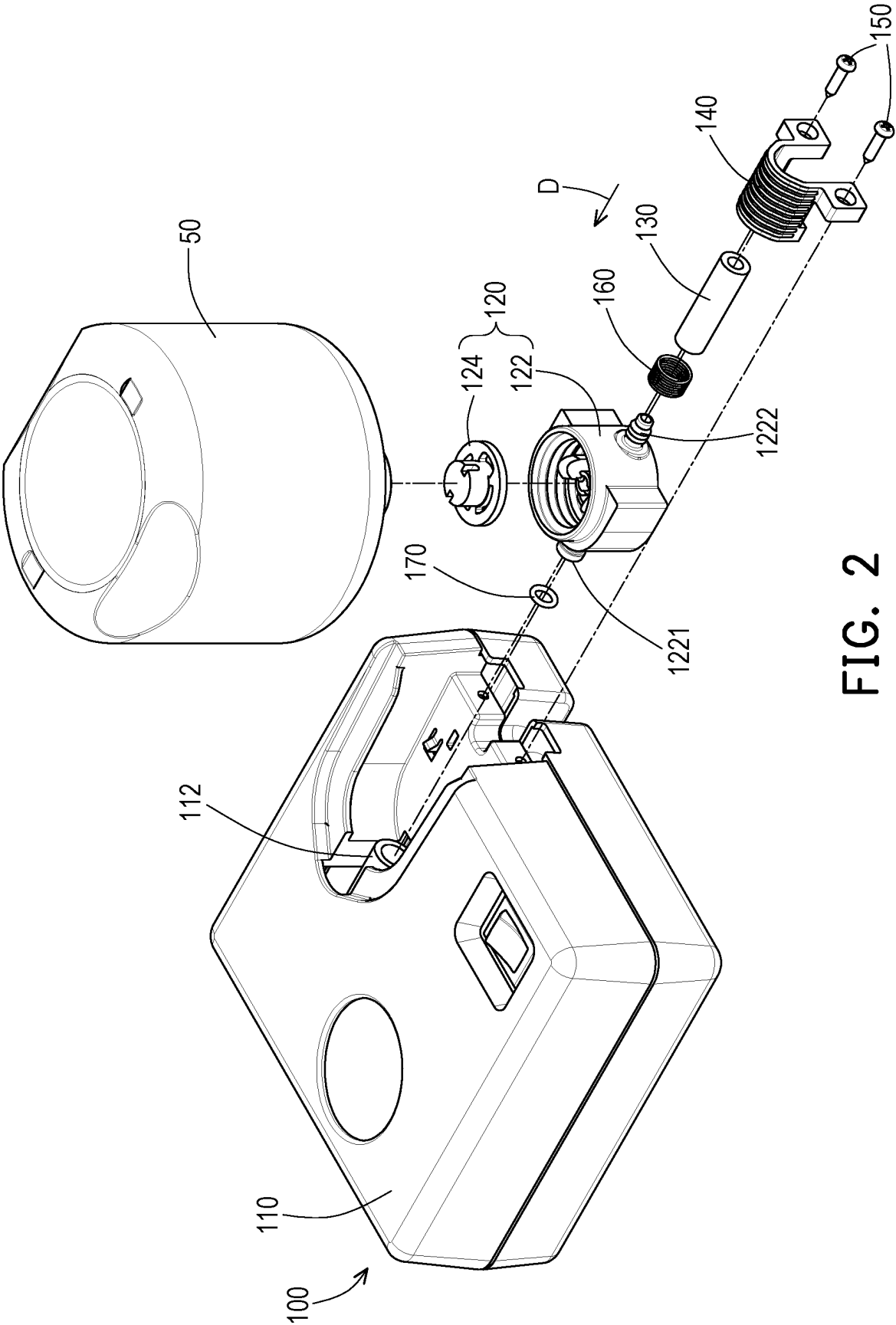


FIG. 2

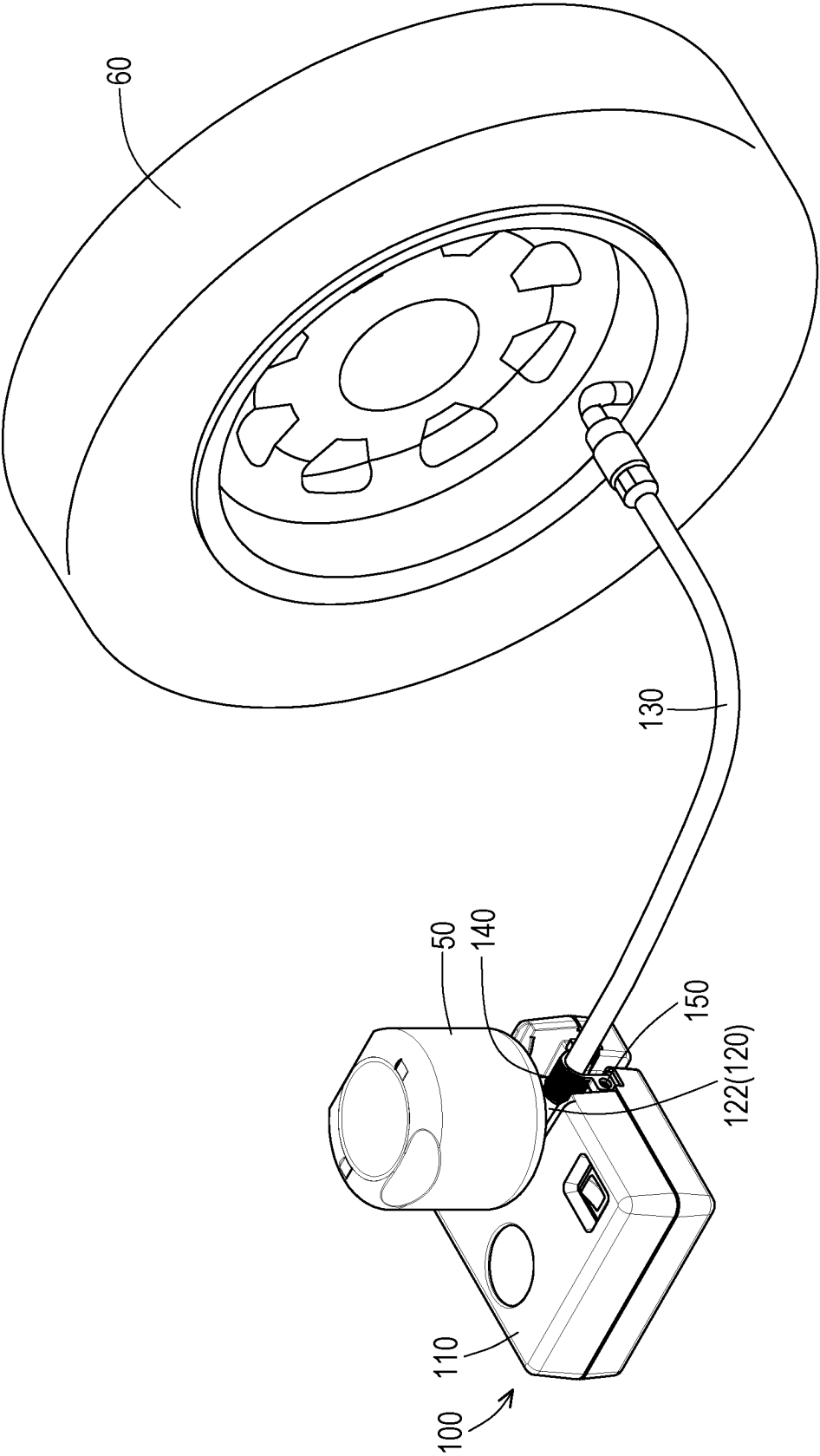


FIG. 3

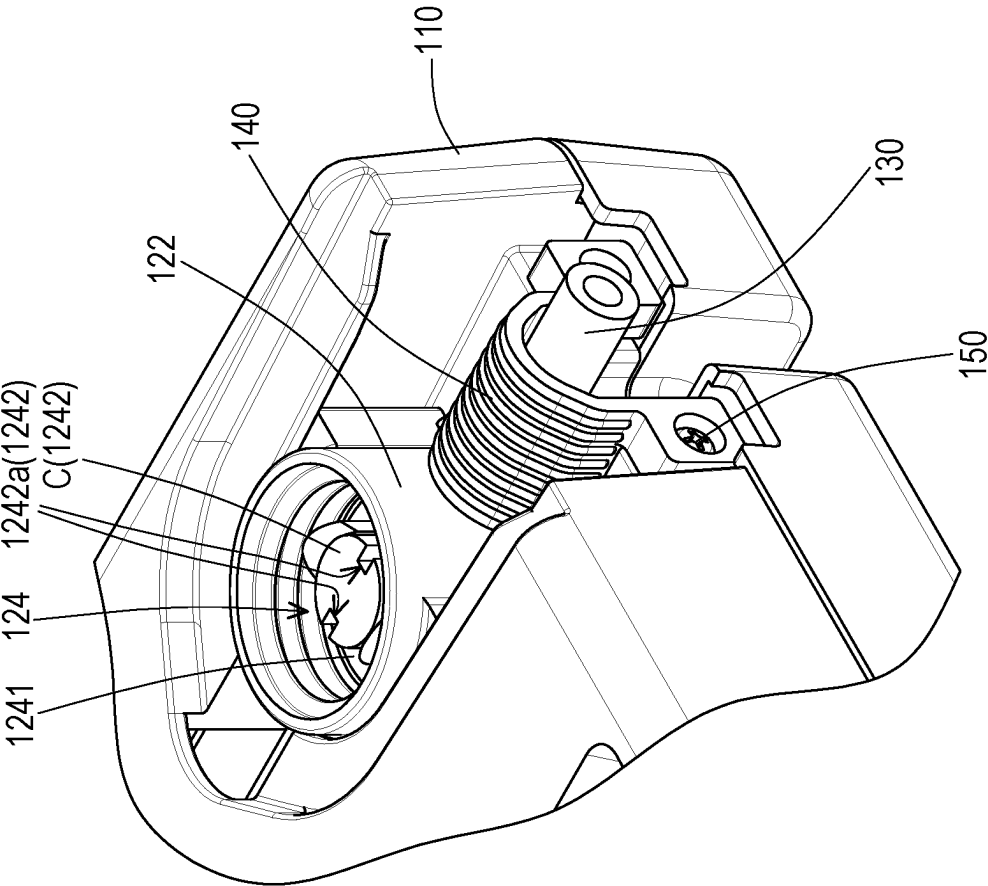


FIG. 4

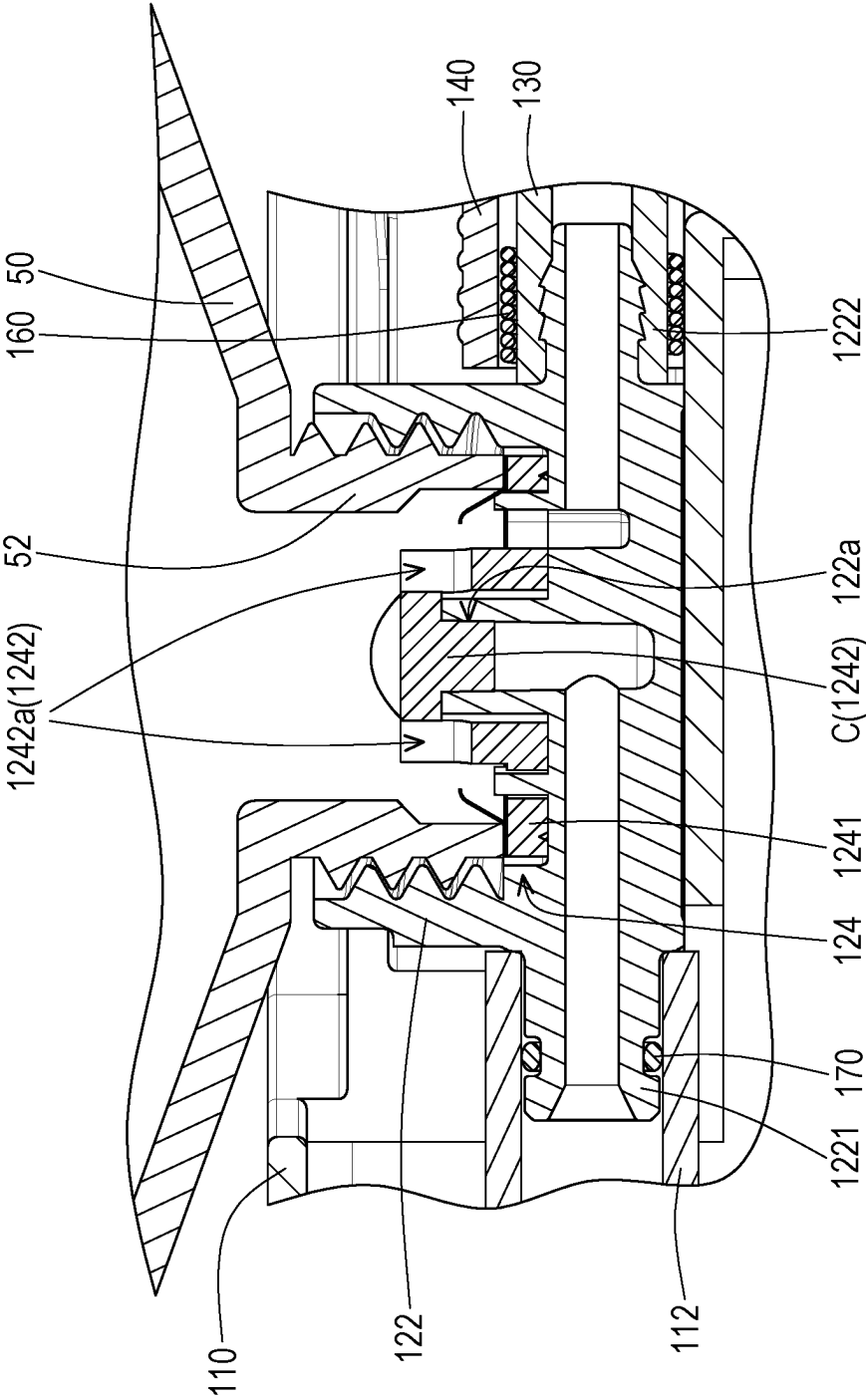


FIG. 5

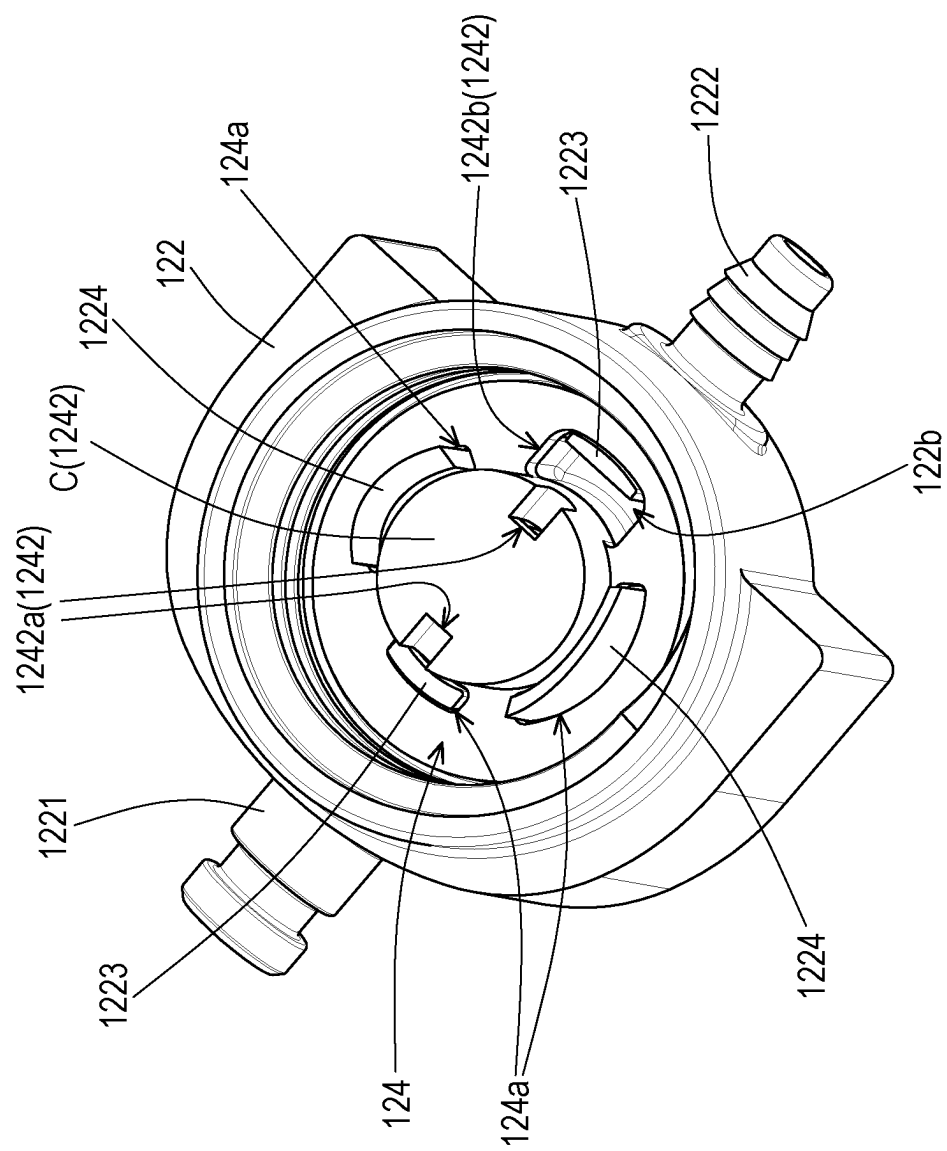


FIG. 6

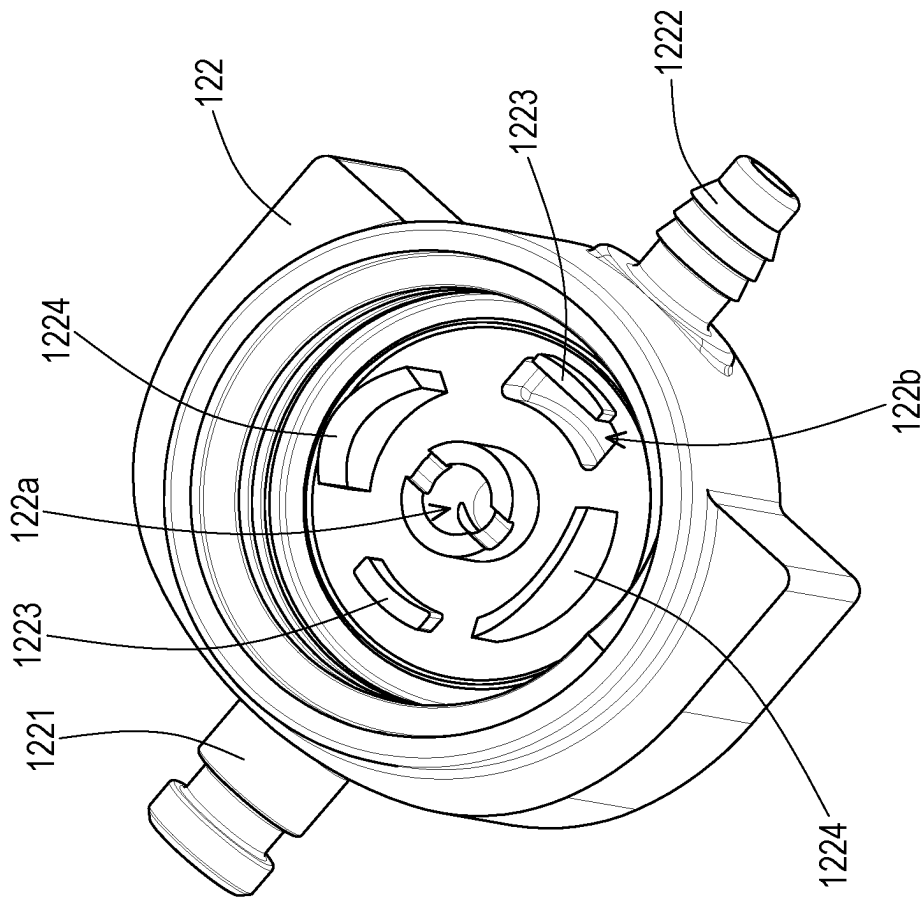


FIG. 7

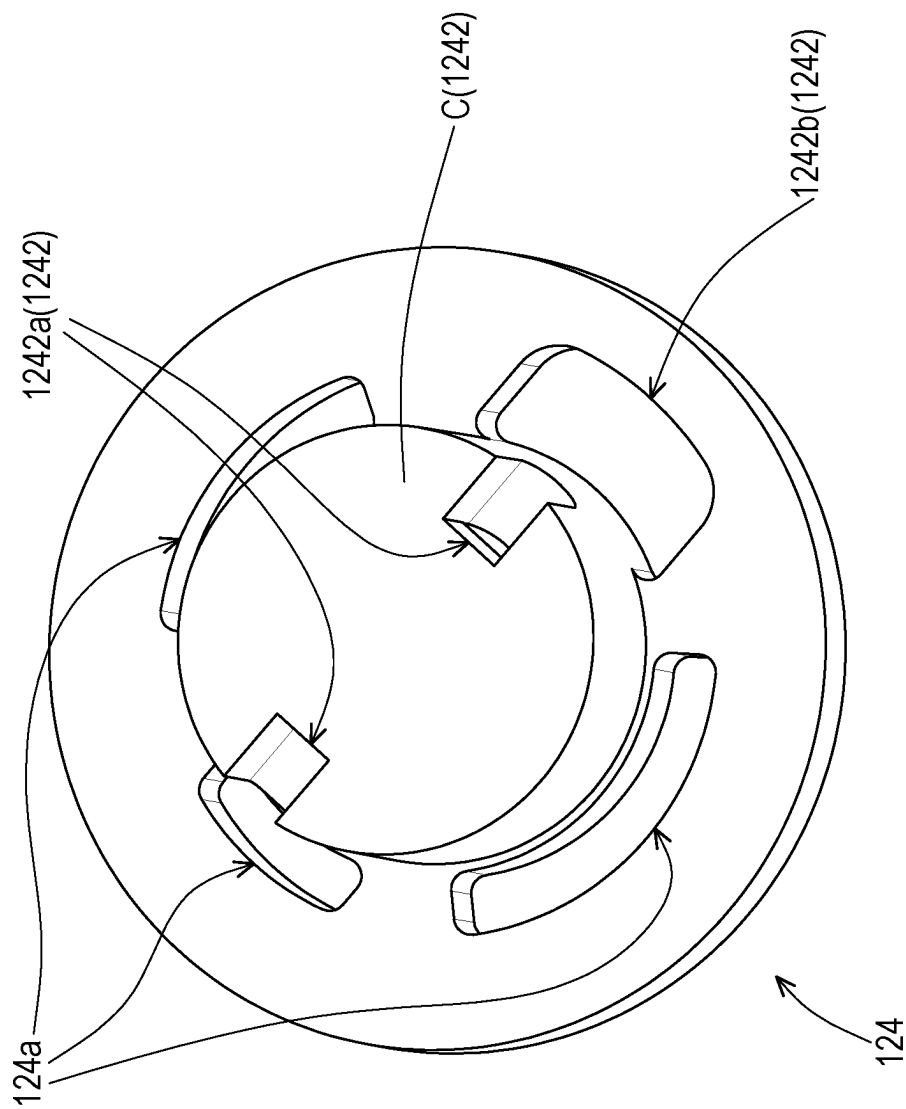


FIG. 8

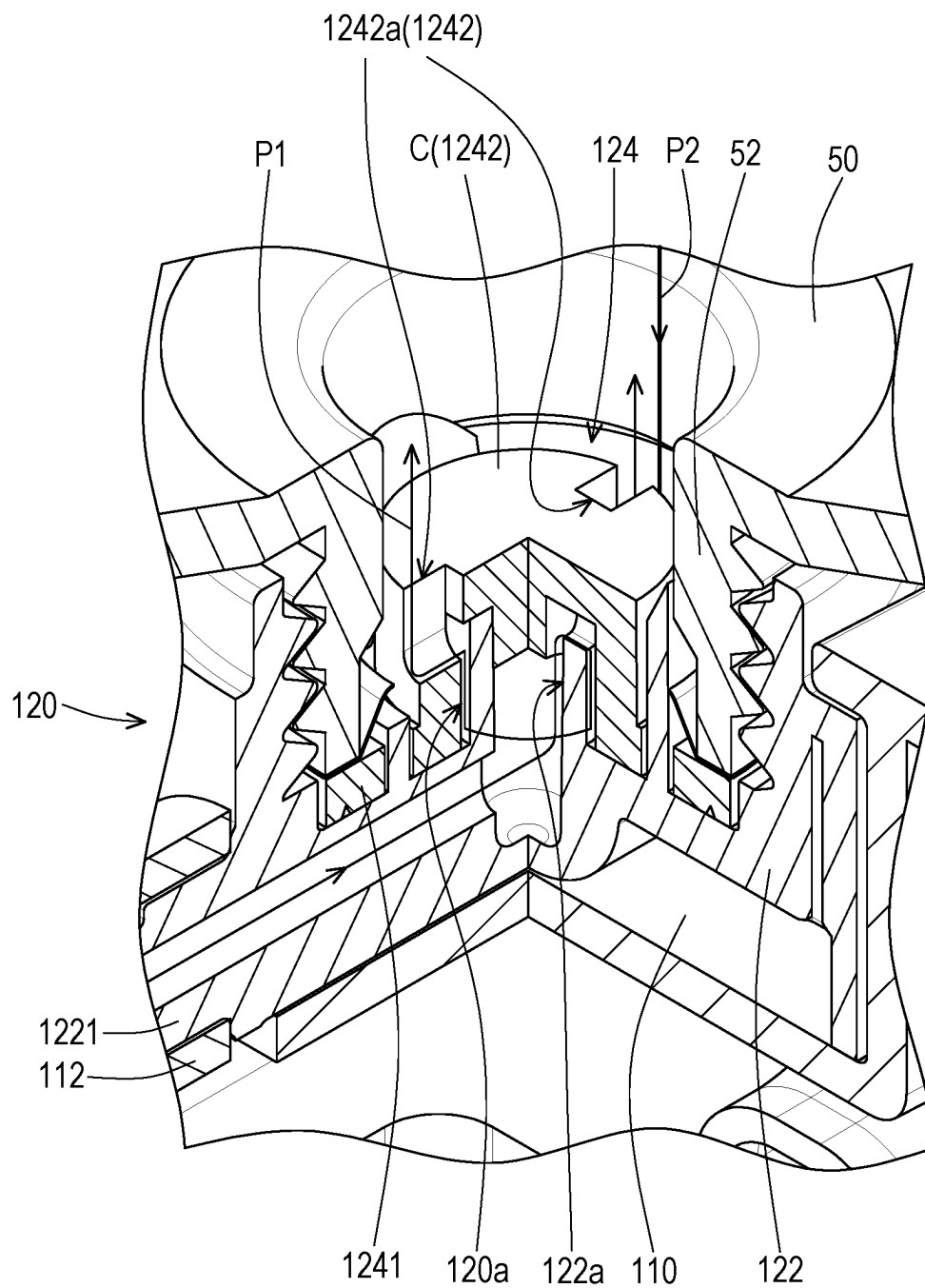


FIG. 9

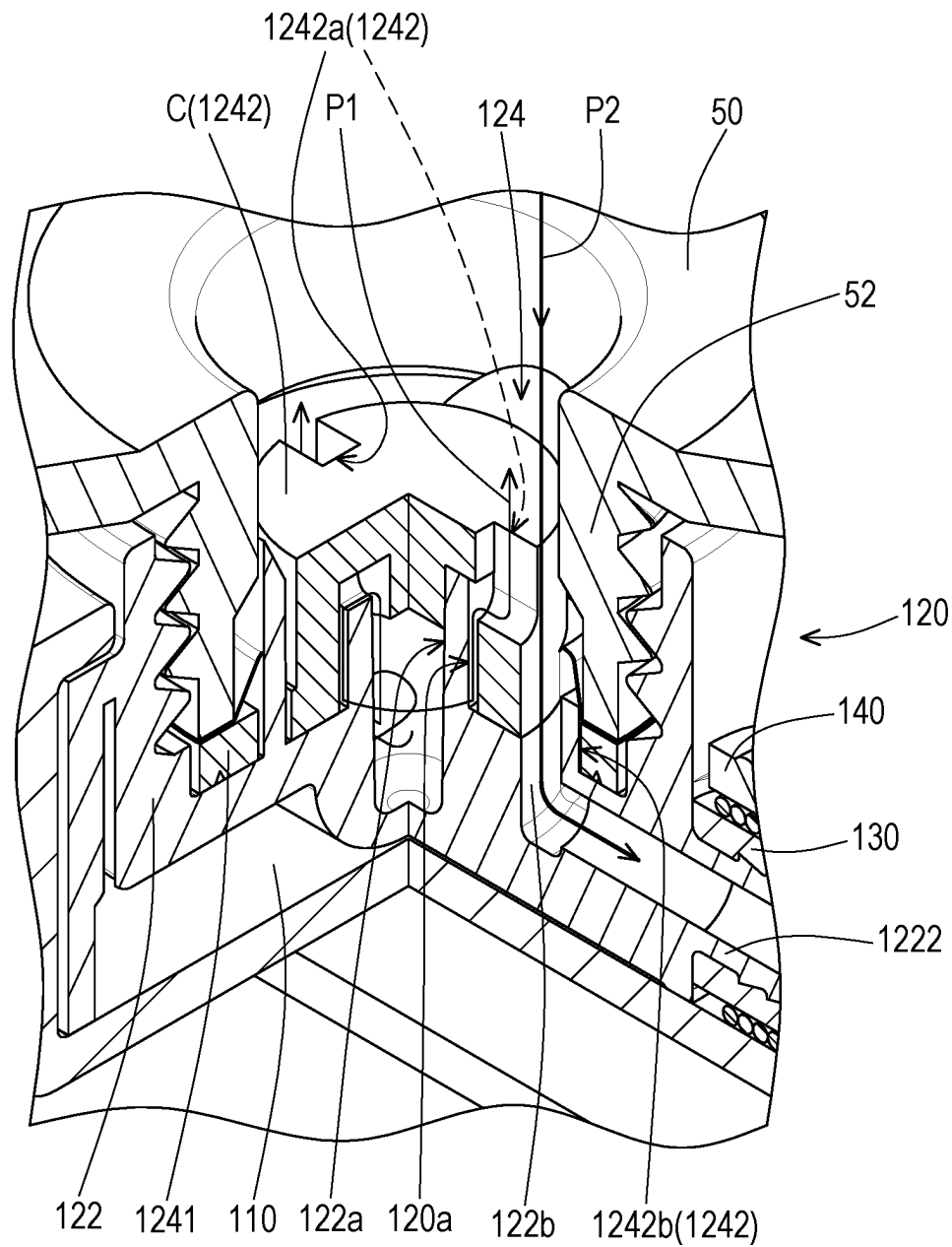


FIG. 10

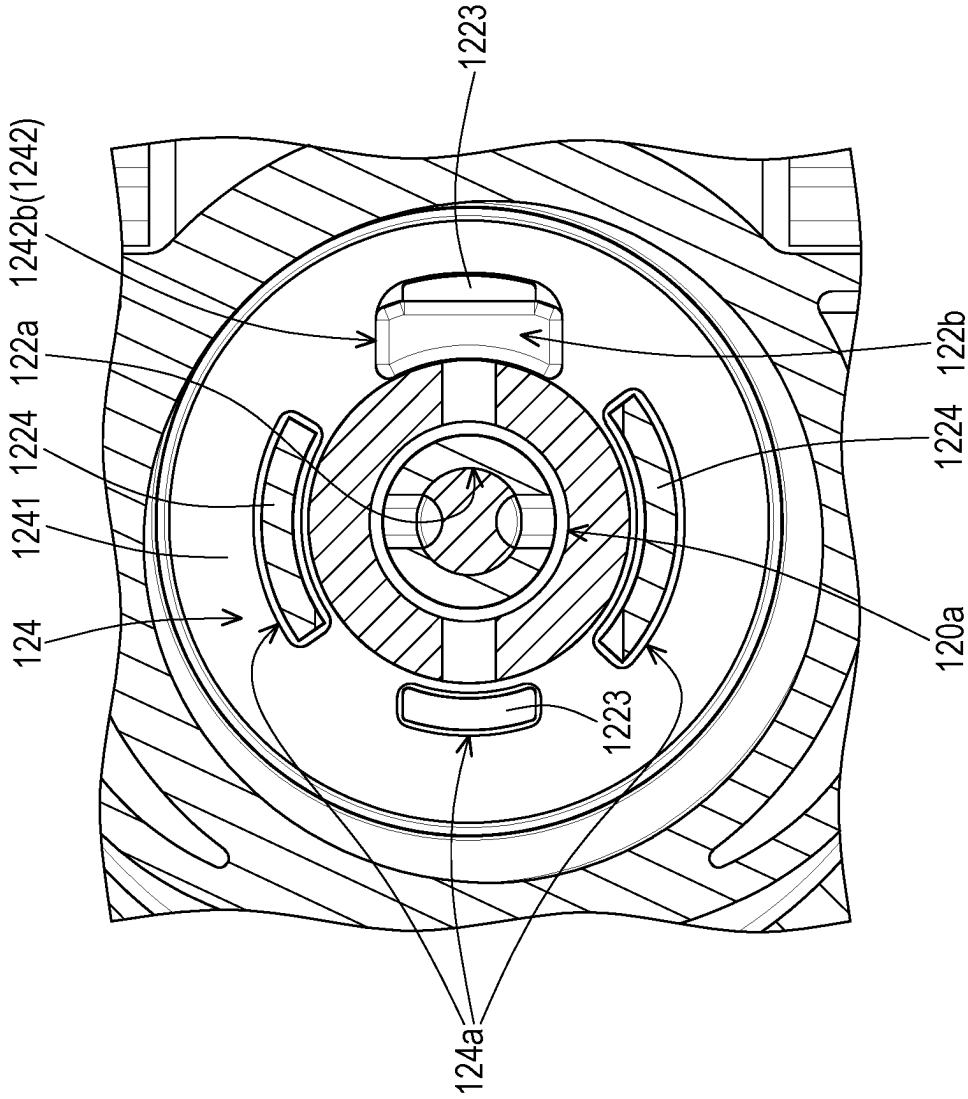


FIG. 11

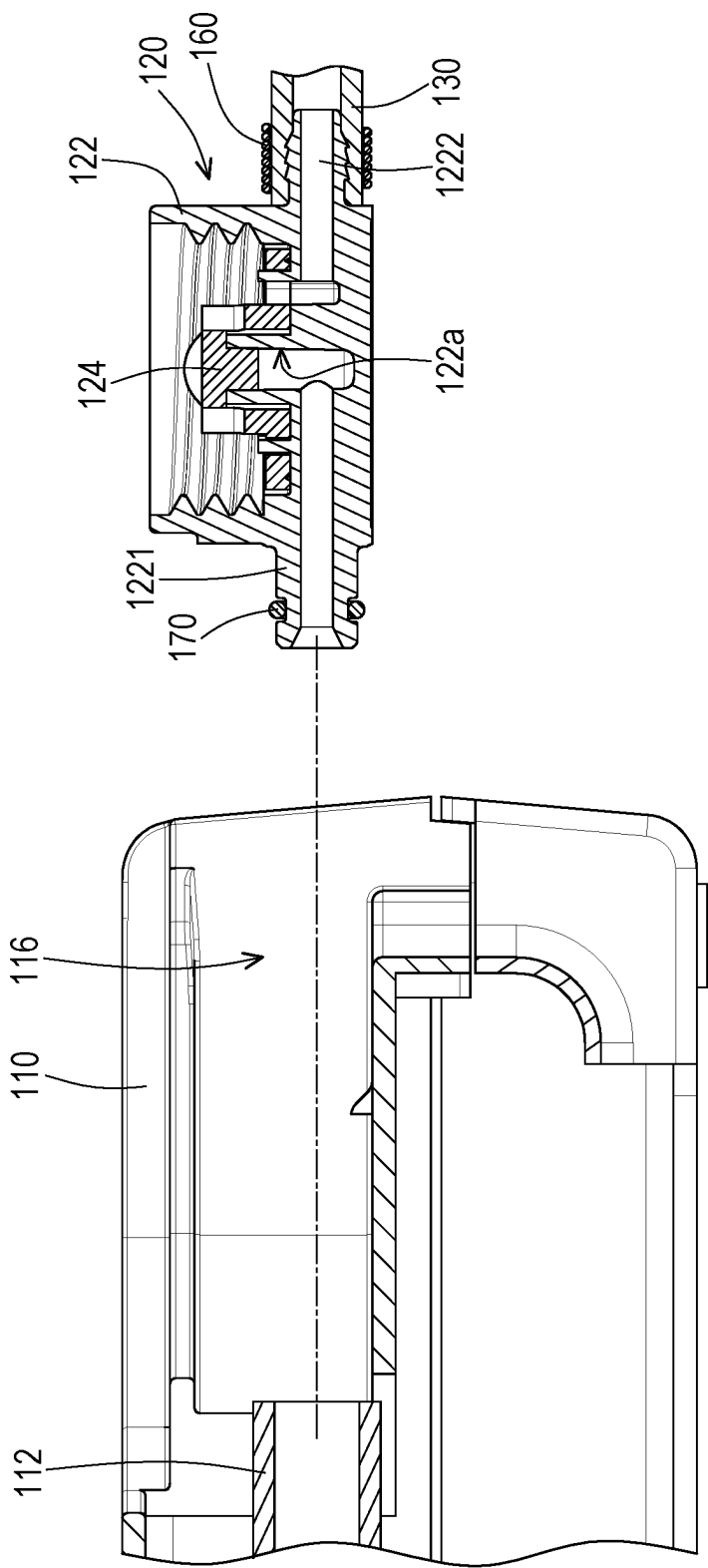


FIG. 12A

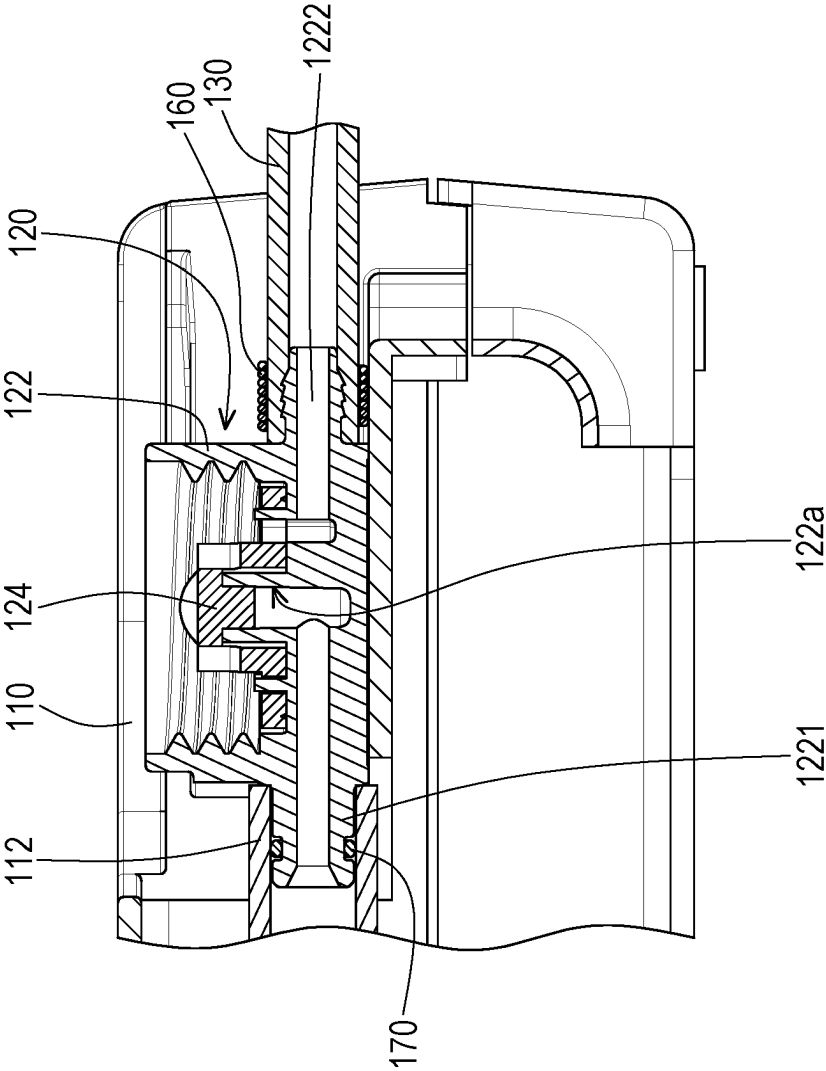


FIG. 12B

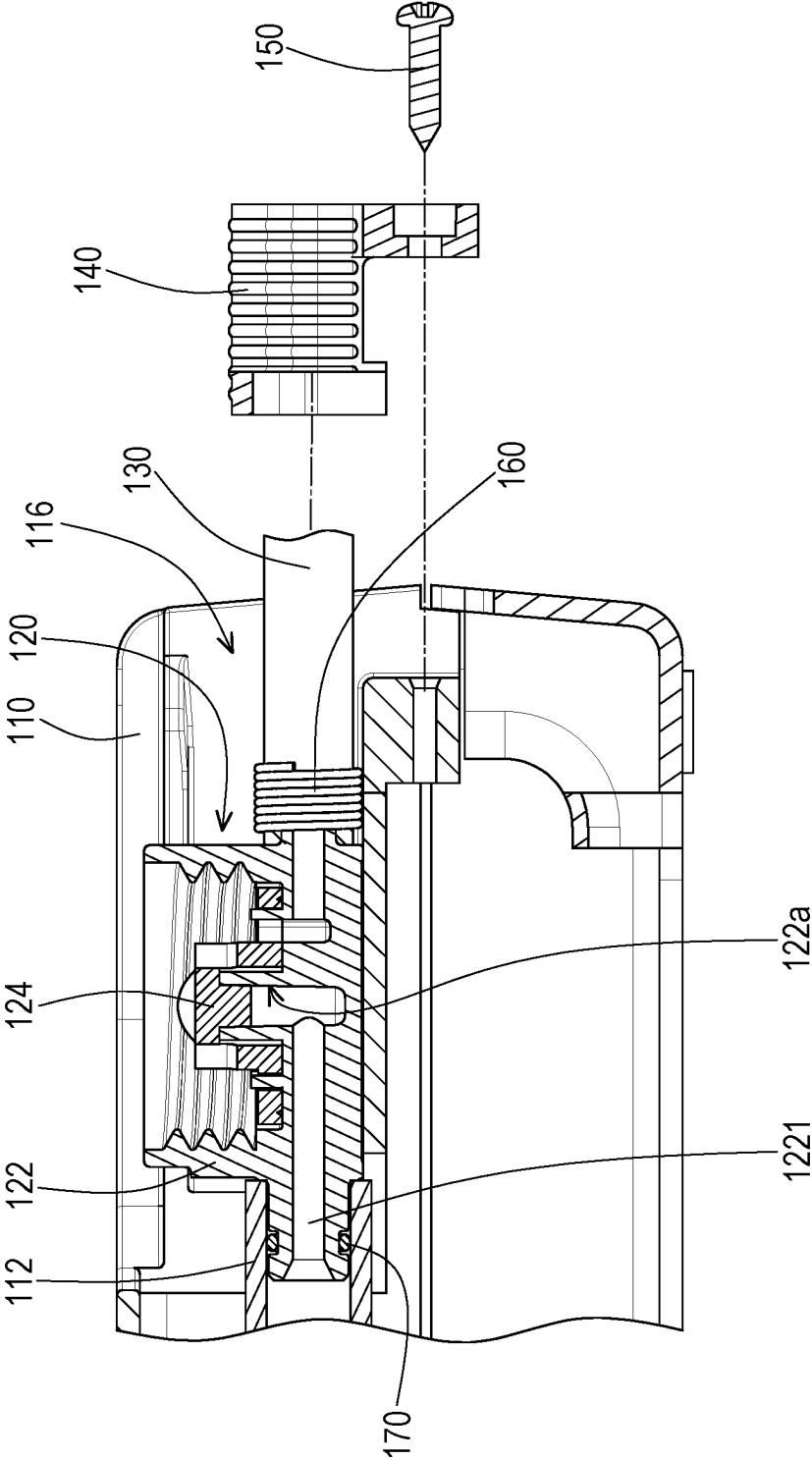


FIG. 12C

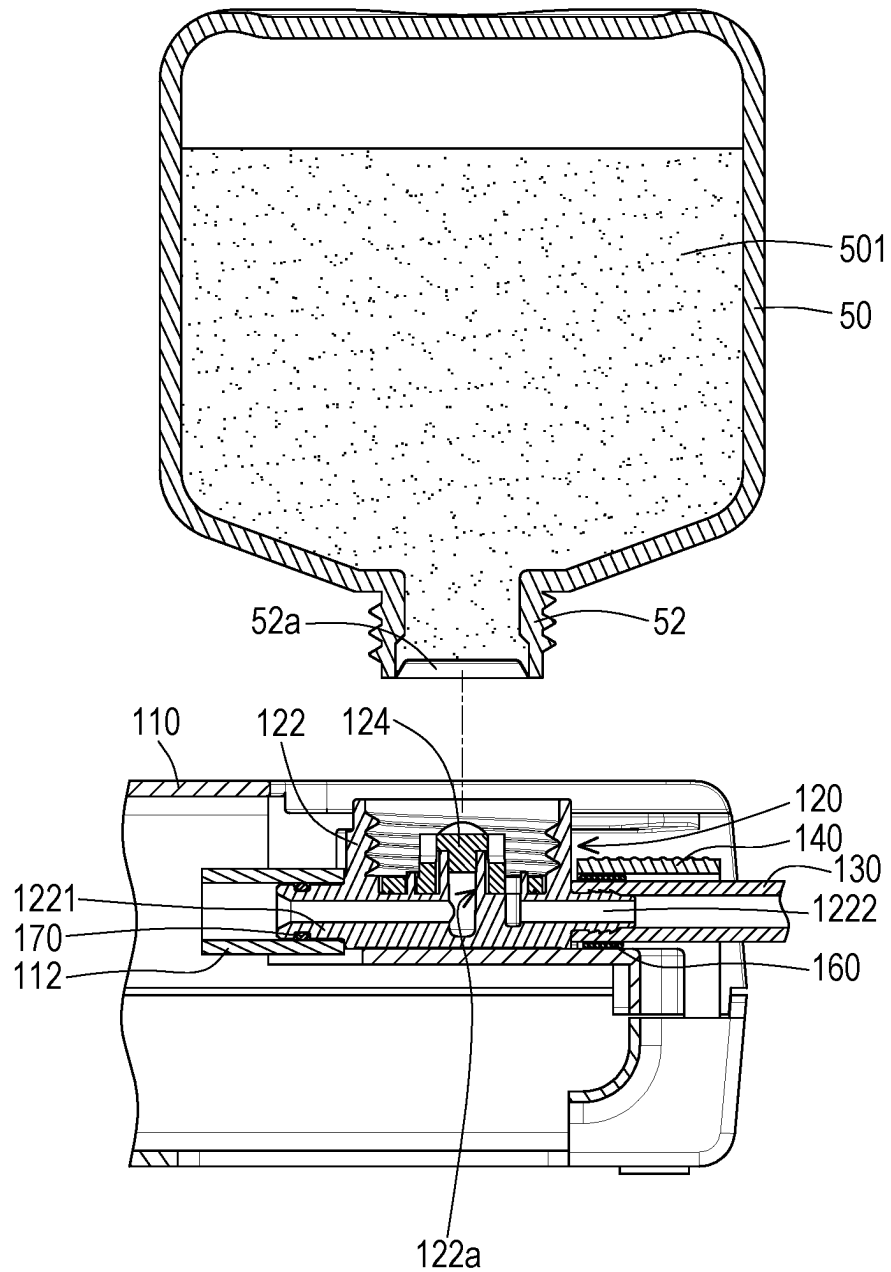


FIG. 12D

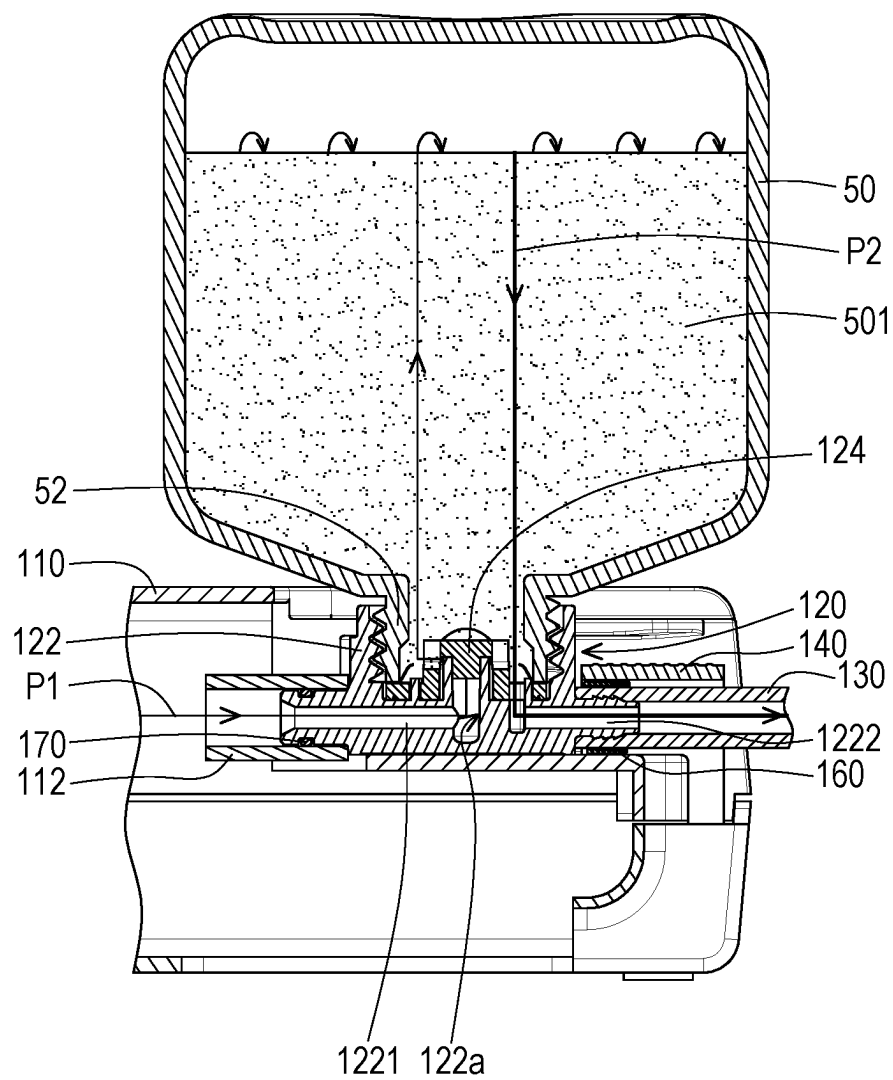


FIG. 12E

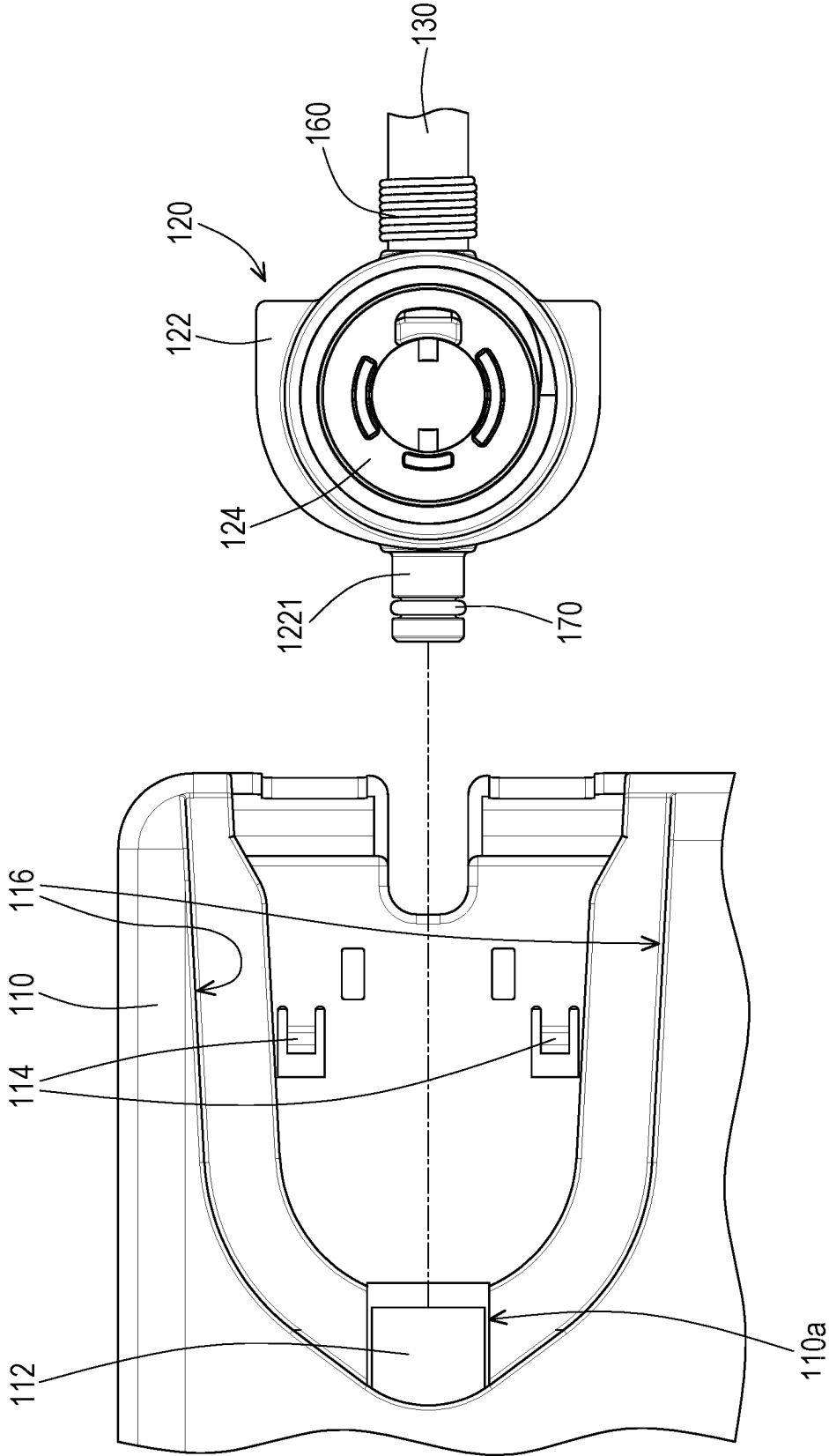


FIG. 13A

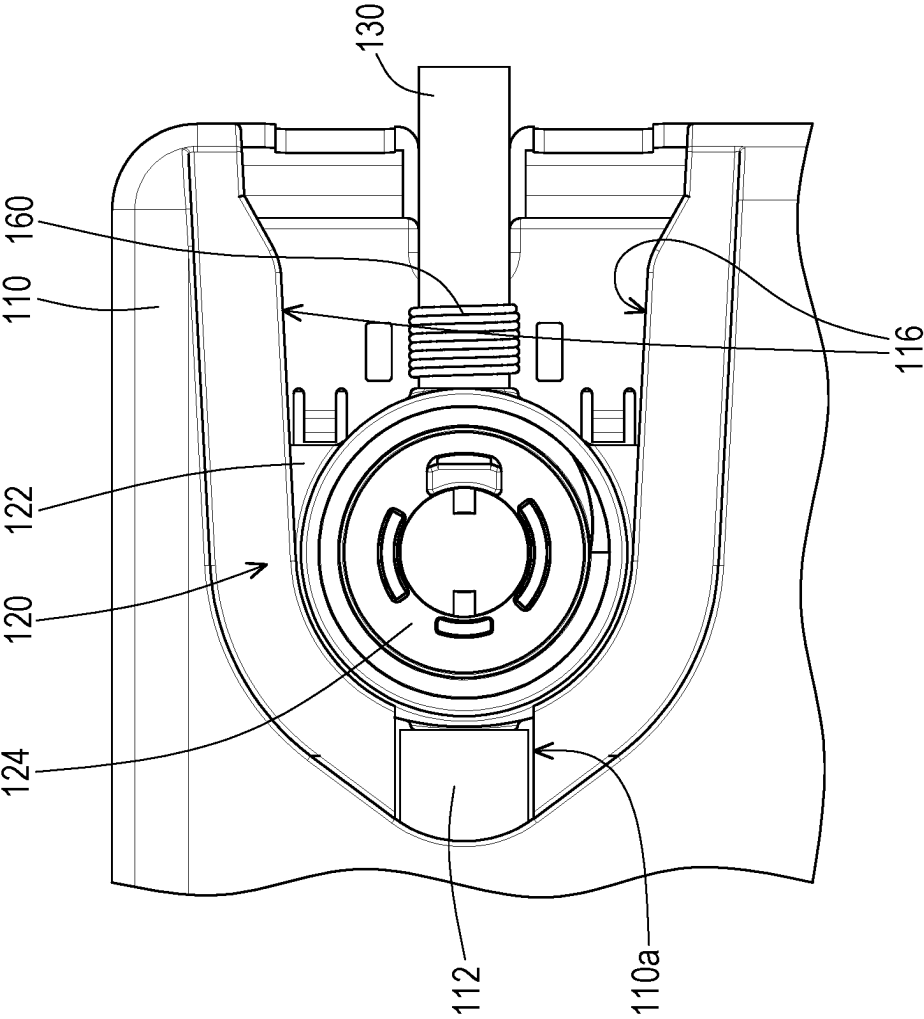


FIG. 13B

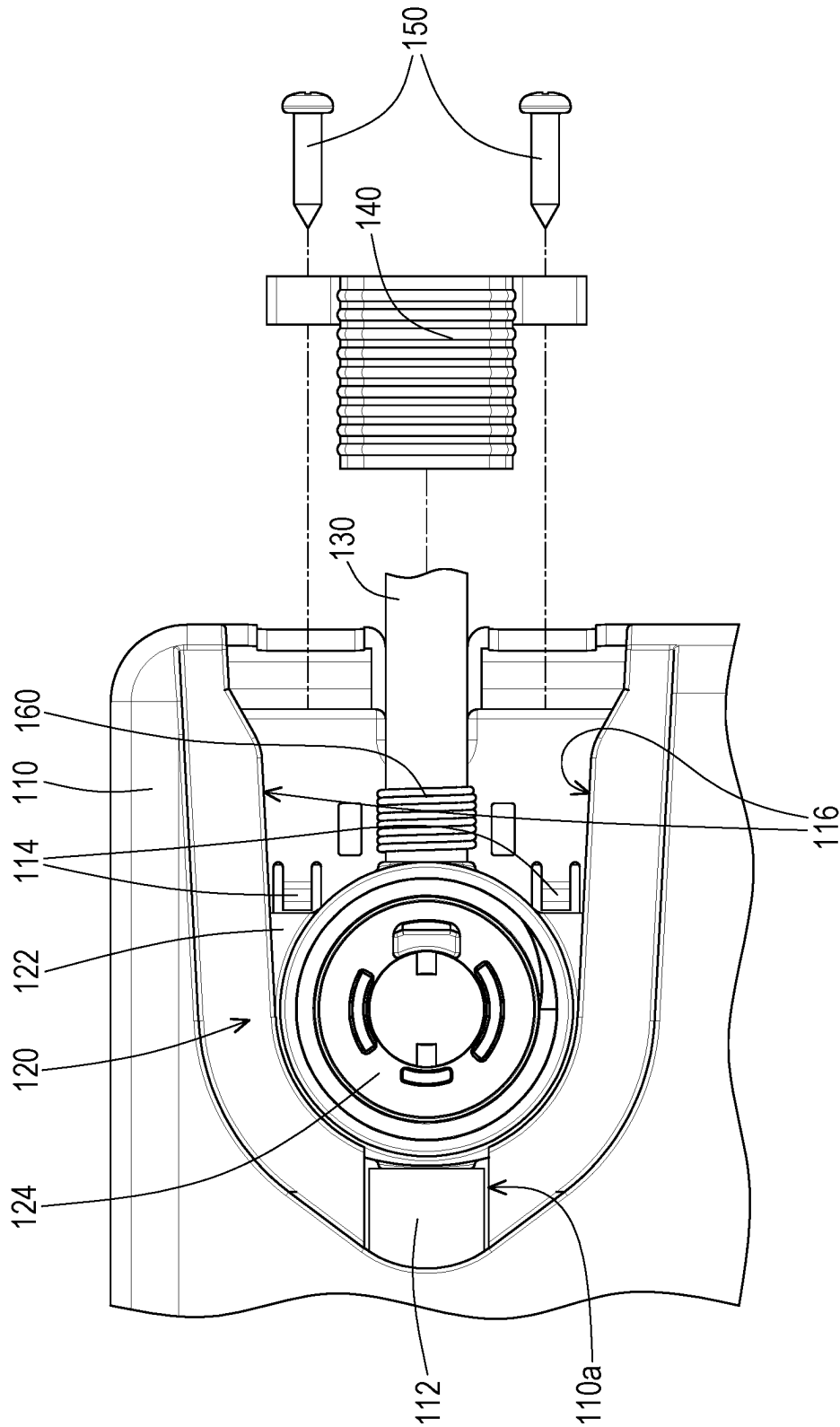


FIG. 13C

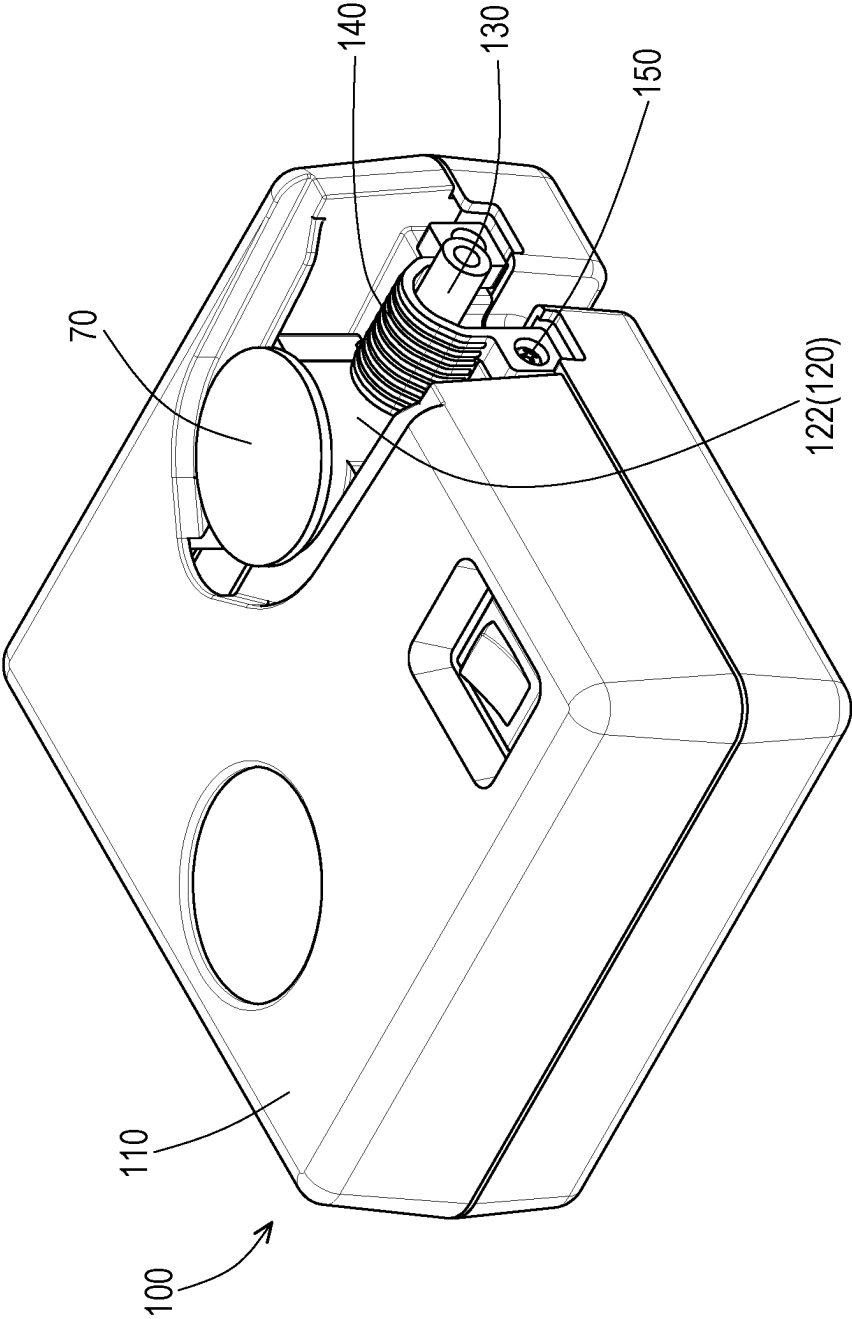


FIG. 14

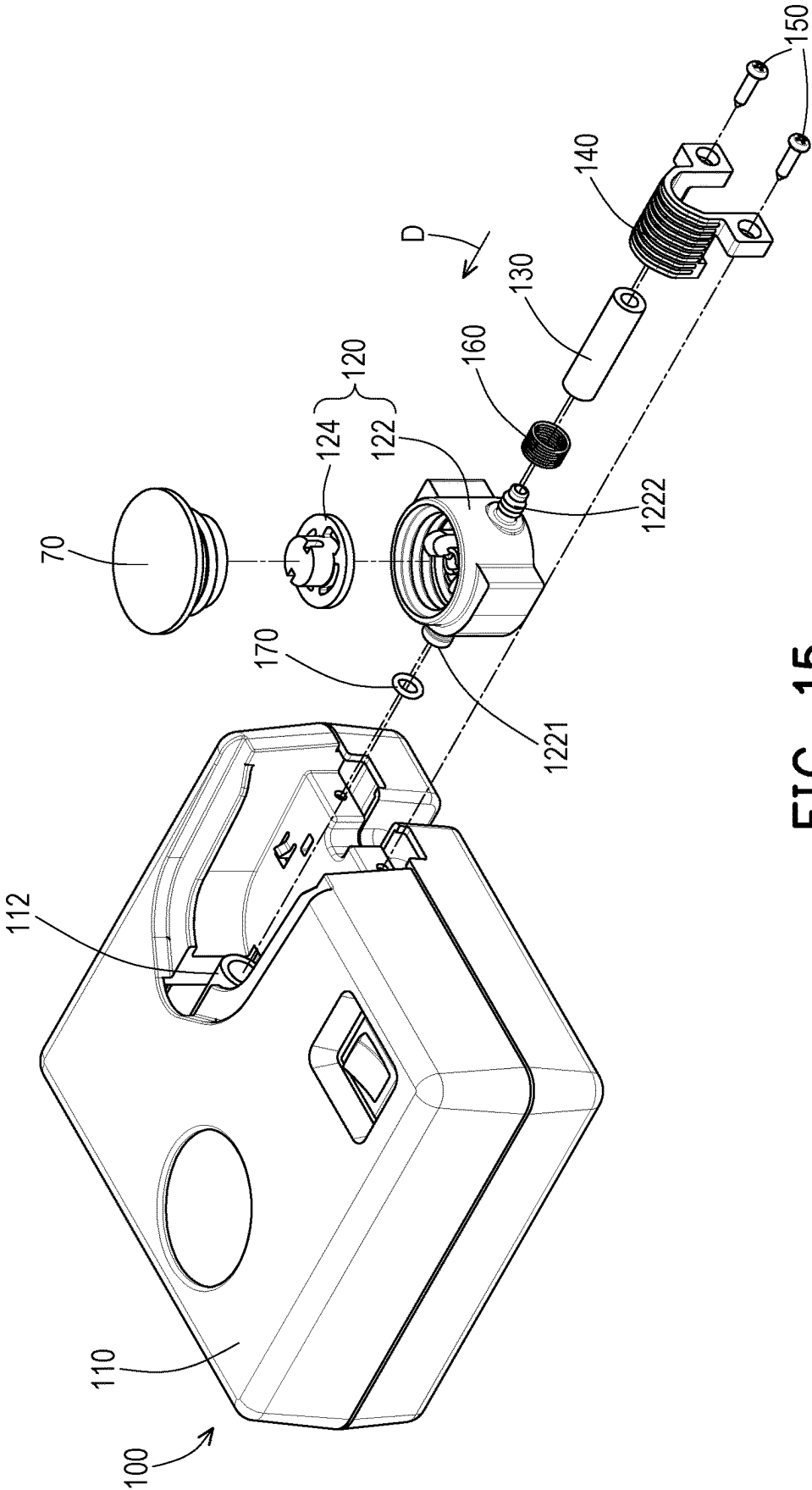


FIG. 15

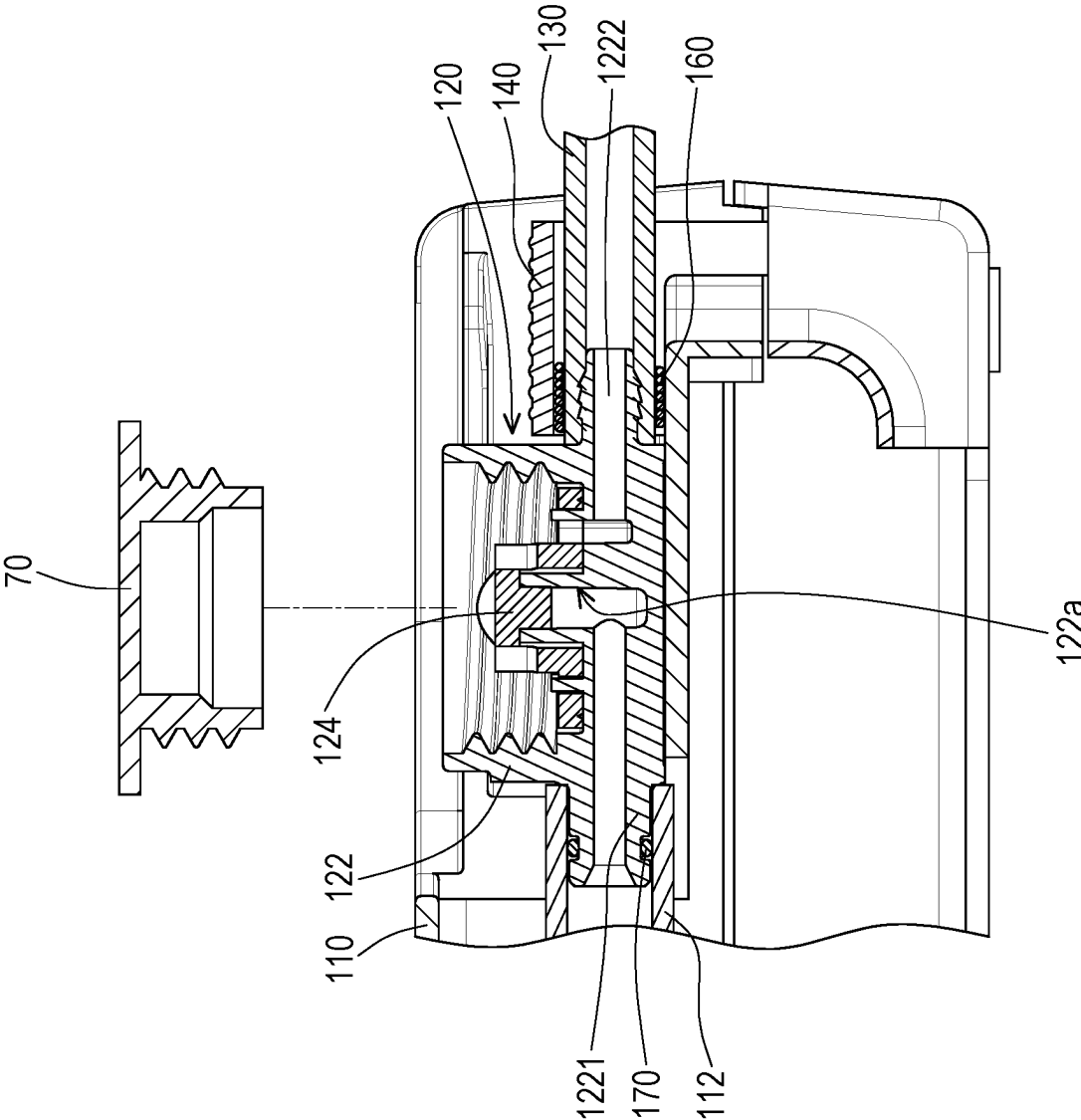


FIG. 16A

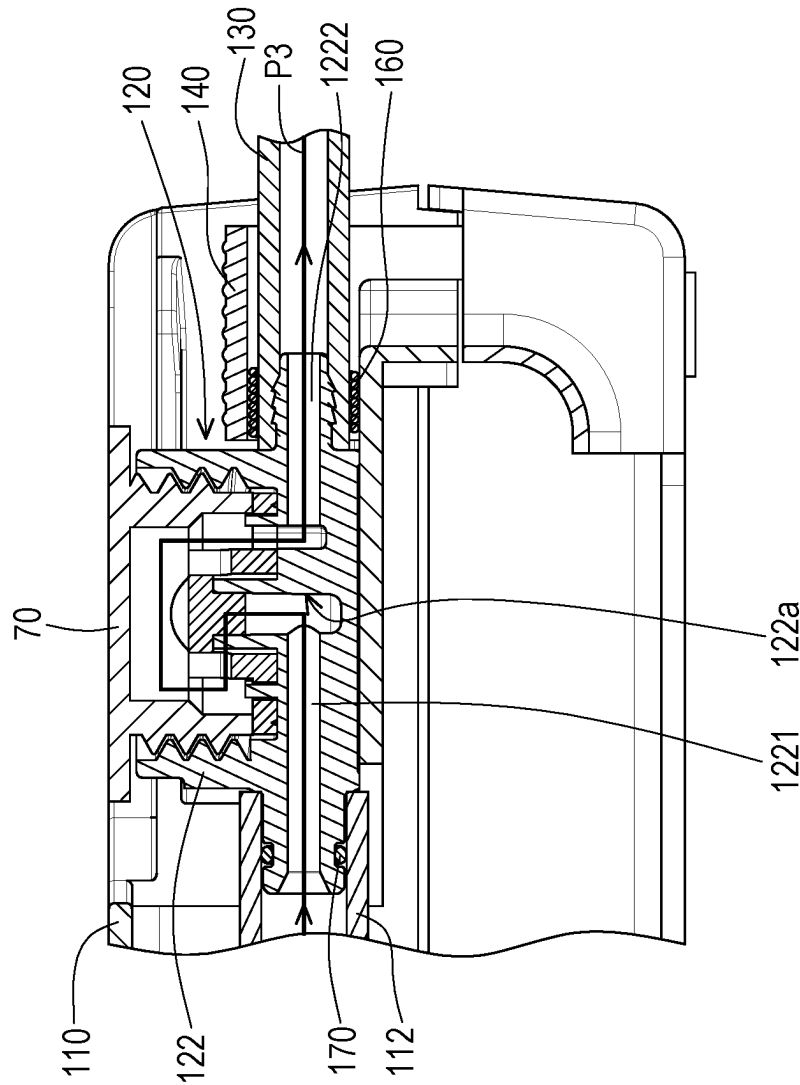


FIG. 16B

VEHICLE-MOUNTED AIR COMPRESSOR**CROSS-REFERENCE TO RELATED APPLICATION**

This application claims the priority benefit of U.S. Provisional Application No. 63/592,553, filed on Oct. 23, 2023 and Taiwan Application No. 113106360, filed on Feb. 22, 2024. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND OF THE INVENTION**Field of the Invention**

The invention relates to an air compressor, and in particular to a vehicle-mounted air compressor.

Description of Related Art

A vehicle-mounted air compressor may be used with a tire sealant holding bottle to repair and inflate the vehicle's tires, and may also be used to inflate the vehicle's tires without a tire sealant holding bottle. In general, when performing a tire repair and inflation operation, the air compressor main body of the vehicle-mounted air compressor needs to be equipped with a bottle cap assembly to combine with the tire sealant holding bottle, and the high-pressure airflow from the air compressor main body passes through the bottle cap assembly and enters the tire sealant holding bottle, and then the high-pressure airflow drives the tire sealant in the tire sealant holding bottle to enter the tire to achieve the effect of tire repair and inflation.

If the tire sealant in the tire sealant holding bottle accidentally flows into the air compressor main body via the bottle cap assembly, the air compressor main body may not operate properly. In order to avoid the above situation, additional parts are added to the bottle cap assembly to prevent the tire sealant from flowing to the air compressor main body, and therefore the number of parts and the difficulty of manufacturing and assembling the bottle cap assembly are increased.

SUMMARY OF THE INVENTION

The invention provides a vehicle-mounted air compressor and a bottle cap assembly thereof that prevent the tire sealant from flowing to the air compressor main body via a simple structure.

A vehicle-mounted air compressor of the invention includes an air compressor main body and a bottle cap assembly. The bottle cap assembly includes a bottle cap component and a sealing and flow guiding component. The bottle cap component is mounted at the air compressor main body and has an air inlet opening and an air outlet opening, and the bottle cap component is connected to the air compressor main body via the air inlet opening. The sealing and flow guiding component is disposed at the bottle cap component and includes a sealing portion and a flow guiding portion. The sealing portion surrounds the flow guiding portion, and the flow guiding portion covers the air inlet opening and has at least one air inlet hole and an air outlet hole. The flow guiding portion is connected to the air inlet opening via the air inlet hole and connected to the air outlet opening via the air outlet hole. An airflow from the air compressor main body is adapted to pass through the air

inlet opening, the air inlet hole, the air outlet hole, and the air outlet opening in sequence.

A bottle cap assembly of the invention is adapted for a vehicle-mounted air compressor and includes a bottle cap component and a sealing guide component. The bottle cap component is mounted at an air compressor main body of the vehicle-mounted air compressor and has an air inlet opening and an air outlet opening, and the bottle cap component is connected to the air compressor main body via the air inlet opening. The sealing and flow guiding component is disposed at the bottle cap component and includes a sealing portion and a flow guiding portion. The sealing portion surrounds the flow guiding portion. The flow guiding portion covers the air inlet opening and has at least one air inlet hole and an air outlet hole. The flow guiding portion is connected to the air inlet opening via the air inlet hole and connected to the air outlet opening via the air outlet hole. An airflow from the air compressor main body is adapted to pass through the air inlet opening, the air inlet hole, the air outlet hole, and the air outlet opening in sequence.

In an embodiment of the invention, the sealing portion and the flow guiding portion are integrally formed structures.

In an embodiment of the invention, a flow guiding channel is formed between the sealing and flow guiding component and the bottle cap component, and the flow guiding channel is connected between the at least one air inlet hole and the air outlet hole.

In an embodiment of the invention, a width of the flow guiding channel is less than an inner diameter of the air inlet opening and an inner diameter of at least one air inlet hole.

In an embodiment of the invention, the flow guiding channel is extended in a zigzag manner.

In an embodiment of the invention, the flow guiding portion includes a boss, the at least one air inlet hole is located at a top end of the boss, and a bottom end of the boss is adjacent to the air outlet hole and the sealing portion.

In an embodiment of the invention, the sealing and flow guiding component has at least one positioning slot, the bottle cap component has at least one positioning rib, and the at least one positioning rib passes through the at least one positioning slot to prevent the sealing and flow guiding component from rotating relative to the bottle cap component.

In an embodiment of the invention, the bottle cap component is adapted to engage with a bottleneck of a tire sealant holding bottle, so that the sealing portion is compressed between the bottle cap component and the bottleneck of the tire sealant holding bottle, and the air inlet hole and the air outlet hole are connected to an internal space of the tire sealant holding bottle.

In an embodiment of the invention, the bottle cap component has at least one cutting edge rib, and the at least one cutting edge rib is adapted to cut a sealing film on the bottleneck of the tire sealant holding bottle.

In an embodiment of the invention, the vehicle-mounted air compressor further includes a limiting component and at least one fastening component, wherein the limiting component limits the bottle cap component to the air compressor main body, and the at least one fastening component fastens the limiting component to the air compressor main body.

In an embodiment of the invention, the air compressor main body has an air outlet pipe, the air outlet pipe is connected to the bottle cap component to be connected to the air inlet opening, there is an opening at a top surface of the

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air compressor main body, and a connection point between the air outlet pipe and the bottle cap component corresponds to the opening.

Based on the above, in the vehicle-mounted air compressor of the invention, in addition to providing the existing sealing function via the sealing portion thereof, the sealing and flow guiding component also covers the air inlet opening of the bottle cap component via the flow guiding portion thereof. Accordingly, while the high-pressure airflow from the air compressor main body flows smoothly to the air outlet opening of the bottle cap component via the guidance of the flow guiding portion, the flow guiding portion may prevent the tire sealant in the tire sealant holding bottle from flowing to the air inlet opening of the bottle cap component. That is, in the invention, the flow guiding portion is integrally formed on the existing sealing component to cover the air inlet opening of the bottle cap component, so no additional parts are needed to achieve this object. Therefore, the vehicle-mounted air compressor and the bottle cap assembly thereof of the invention prevent the tire sealant from flowing to the air compressor main body via a simple structure, thereby reducing the difficulty of manufacturing and assembling the bottle cap assembly.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a three-dimensional view of a vehicle-mounted air compressor and a tire sealant holding bottle of an embodiment of the invention.

FIG. 2 is an exploded view of the vehicle-mounted air compressor and the tire sealant holding bottle of FIG. 1.

FIG. 3 is a three-dimensional view of the vehicle-mounted air compressor of FIG. 1 connected to a tire.

FIG. 4 is a partial three-dimensional view of the vehicle-mounted air compressor of FIG. 1.

FIG. 5 is a partial cross-sectional view of the vehicle-mounted air compressor and the tire sealant holding bottle of FIG. 1.

FIG. 6 is a three-dimensional view of the bottle cap component and the sealing and flow guiding component of FIG. 2.

FIG. 7 is a three-dimensional view of the bottle cap component of FIG. 6.

FIG. 8 is a three-dimensional view of the sealing and flow guiding component of FIG. 6.

FIG. 9 shows the partial structure of the vehicle-mounted air compressor and the tire sealant holding bottle of FIG. 1.

FIG. 10 shows the partial structure of the vehicle-mounted air compressor and the tire sealant holding bottle of FIG. 1.

FIG. 11 shows the partial structure of the bottle cap component and the sealing and flow guiding component of FIG. 6.

FIG. 12A to FIG. 12E are operation flowcharts of the vehicle-mounted air compressor of FIG. 1.

FIG. 13A to FIG. 13C are top views of the vehicle-mounted air compressor of FIG. 12A to FIG. 12C respectively.

FIG. 14 shows that the tire sealant holding bottle in FIG. 1 is replaced by a cover body.

FIG. 15 shows that the tire sealant holding bottle in FIG. 2 is replaced by a cover body.

FIG. 16A shows that the tire sealant holding bottle in FIG. 12D is replaced by a cover body.

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FIG. 16B shows that the tire sealant holding bottle in FIG. 12E is replaced by a cover body.

DESCRIPTION OF THE EMBODIMENTS

FIG. 1 is a three-dimensional view of a vehicle-mounted air compressor and a tire sealant holding bottle of an embodiment of the invention. FIG. 2 is an exploded view of the vehicle-mounted air compressor and the tire sealant holding bottle of FIG. 1. FIG. 3 is a three-dimensional view of the vehicle-mounted air compressor of FIG. 1 connected to a tire. Please refer to FIG. 1 to FIG. 3. A vehicle-mounted air compressor 100 of the present embodiment includes an air compressor main body 110, a bottle cap assembly 120, and a connecting pipe 130. The bottle cap assembly 120 includes a bottle cap component 122 and a sealing and flow guiding component 124. The bottle cap component 122 is mounted at the air compressor main body 110 and has an air inlet end 1221 and an air outlet end 1222. The air inlet end 1221 and the air outlet end 1222 are, for example, cylindrical bodies protruded outward from the outer surface of the bottle cap component 122. The air inlet end 1221 is adapted to be plugged into an air outlet pipe 112 of the air compressor main body 110. The sealing and flow guiding component 124 is made of, for example, rubber, silicone, or other types of elastic sealing materials and is disposed at the bottle cap component 122. The tire sealant holding bottle 50 is used to be engaged with the bottle cap component 122. The bottle cap component 122 may be connected to a tire 60 via the connecting pipe 130.

The high-pressure airflow from the air compressor main body 110 is adapted to enter the bottle cap component 122 via the air inlet end 1221 of the bottle cap component 122 and pass through the bottle cap component 122 and the sealing and flow guiding component 124 in sequence to reach the tire sealant holding bottle 50. Then, the tire sealant in the tire sealant holding bottle 50 is driven via the high-pressure airflow to pass through the sealing and flow guiding component 124 and the bottle cap component 122 in sequence and leave the bottle cap component 122 via the air outlet end 1222 of the bottle cap component 122 and flow to the tire 60 through the connecting pipe 130 to achieve the effect of tire repair and inflation. The method and principle by which the air compressor main body 110 generates high-pressure airflow are known in the technical field and are not described in detail here.

FIG. 4 is a partial three-dimensional view of the vehicle-mounted air compressor of FIG. 1. FIG. 5 is a partial cross-sectional view of the vehicle-mounted air compressor and the tire sealant holding bottle of FIG. 1. Please refer to FIG. 4 and FIG. 5. Specifically, the sealing and flow guiding component 124 is located in the bottle cap component 122 and includes a sealing portion 1241 and a flow guiding portion 1242. The sealing portion 1241 surrounds the flow guiding portion 1242, and the flow guiding portion 1242 is connected between the air inlet end 1221 and the air outlet end 1222. The sealing portion 1241 and the flow guiding portion 1242 are integrally formed structures. That is, the sealing and flow guiding portion 124 is an integrally formed single member rather than being formed by multiple parts. When the bottleneck 52 of the tire sealant holding bottle 50 is engaged with the bottle cap component 122, the bottleneck 52 of the tire sealant holding bottle 50 presses down the sealing portion 1241 of the sealing and flow guiding component 124, so that the sealing portion 1241 is compressed between the bottle cap component 122 and the bottleneck 52 of the tire sealant holding bottle 50 to prevent the tire sealant

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in the tire sealant holding bottle **50** from flowing out via the gap between the bottleneck **52** and the bottle cap component **122**.

FIG. **6** is a three-dimensional view of the bottle cap component and the sealing and flow guiding component of FIG. **2**. FIG. **7** is a three-dimensional view of the bottle cap component of FIG. **6**. FIG. **8** is a three-dimensional view of the sealing and flow guiding component of FIG. **6**. Please refer to FIG. **6** to FIG. **8**, specifically, the bottle cap component **122** of the present embodiment has an air inlet opening **122a** and an air outlet opening **122b**. The air inlet opening **122a** is connected to the air inlet end **1221**, and the air outlet opening **122b** is connected to the air outlet end **1222**. Therefore, the inside of the bottle cap component **122** may be connected to the air compressor main body **110** via the air inlet opening **122a** and may be connected to the connecting pipe **130** via the air outlet opening **122b**. Moreover, the air guiding portion **1242** covers the air inlet opening **122a** and has at least one air inlet hole **1242a** (two are shown) and an air outlet hole **1242b**. The flow guiding portion **1242** is connected to the air inlet opening **122a** via the air inlet hole **1242a** and connected to the air outlet opening **122b** via the air outlet hole **1242b**. When the bottle cap component **122** and the bottleneck **52** of the tire sealant holding bottle **50** are engaged with each other as shown in FIG. **5**, the air inlet hole **1242a** and the air outlet hole **1242b** of the flow guiding portion **1242** are connected to the internal space of the tire sealant holding bottle **50**. The high-pressure airflow from the air compressor main body **110** is adapted to pass through the air inlet opening **122a**, the air inlet hole **1242a**, the internal space of the tire sealant holding bottle **50**, the air outlet hole **1242b**, and the air outlet opening **122b** in sequence.

Under the above arrangement, the sealing and flow guiding component **124** not only provides the existing sealing function with the sealing portion **1241** thereof, but also covers the air inlet opening **122a** of the bottle cap component **122** via the flow guiding portion **1242** thereof. Accordingly, while the high-pressure airflow from the air compressor main body **110** flows smoothly to the air outlet opening **122b** of the bottle cap component **122** via the guidance of the flow guiding portion **1242**, the flow guiding portion **1242** may prevent the tire sealant in the tire sealant holding bottle **50** from flowing to the air inlet opening **122a** of the bottle cap component **122**. That is, in the present embodiment, the flow guiding portion **1242** is integrally formed on the existing sealing component to cover the air inlet opening **122a** of the bottle cap component **122**, so no additional parts are needed to achieve this object. Therefore, the vehicle-mounted air compressor **100** and the bottle cap assembly **120** thereof of the present embodiment prevent the tire sealant from flowing to the air compressor main body **110** via a simple structure, thereby reducing the difficulty of manufacturing and assembling the bottle cap assembly **120**.

FIG. **9** shows the partial structure of the vehicle-mounted air compressor and the tire sealant holding bottle of FIG. **1**. FIG. **10** shows the partial structure of the vehicle-mounted air compressor and the tire sealant holding bottle of FIG. **1**. FIG. **11** shows the partial structure of the bottle cap component and the sealing and flow guiding component of FIG. **6**. Please refer to FIG. **9** to FIG. **11**, more specifically, in the present embodiment, a flow guiding channel **120a** is formed in the gap between the sealing and flow guiding component **124** and the bottle cap component **122**, and the flow guiding channel **120a** is connected between the air inlet hole **1242a** and the air outlet hole **1242b**. The high-pressure airflow from the air compressor main body **110** (shown in FIG. **1**)

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may pass through the air inlet end **1221**, the air inlet opening **122a**, the flow guiding channel **120a**, and the air inlet hole **1242a** in sequence along a flow path **P1** shown in FIG. **9** to reach the inside of the tire sealant holding bottle **50**, and a tire sealant **501** inside the tire sealant holding bottle **50** may pass through the air outlet hole **1242b**, the air outlet opening **122b**, and the air outlet end **1222** in sequence along a flow path **P2** shown in FIG. **10** with the high-pressure airflow and flow to the tire **60** (shown in FIG. **3**) via the connecting pipe **130**.

More specifically, the flow guiding channel **120a** is extended in a zigzag manner as shown in FIG. **9** and FIG. **10**, and the width of the flow guiding channel **120a** (that is, the gap between the sealing and flow guiding component **124** and the bottle cap component **122**) is less than the inner diameter of the air inlet opening **122a** and the inner diameter of the air inlet hole **1242a**. Accordingly, it may be ensured that the tire sealant having surface tension in the tire sealant holding bottle **50** does not flow to the air inlet opening **122a** of the bottle cap component **122** via the flow guiding channel **120a**. The width of the flow guiding channel **120a** is, for example, 0.1 mm to 0.3 mm, but the invention is not limited thereto.

Please refer to FIG. **8**. In the present embodiment, the flow guiding portion **1242** of the sealing and flow guiding component **124** includes a boss **C**, the air inlet hole **1242a** is located at the top end of the boss **C**, and the bottom end of the boss **C** is adjacent to the air outlet hole **1242b** and the sealing portion **1241**. That is, the flow guiding portion **1242** of the sealing and flow guiding component **124** includes the boss **C**, the air inlet hole **1242a** formed at the top end of the boss **C**, and the air outlet hole **1242b** adjacent to the bottom end of the boss **C**. The air inlet opening **122a** and the flow guiding channel **120a** are located inside the boss **C**. When the tire sealant holding bottle **50** is engaged with the bottle cap component **122**, the boss **C** is protruded into the tire sealant holding bottle **50** and makes the air inlet hole **1242a** reliably face the internal space of the tire sealant holding bottle **50**.

Please refer to FIG. **6** to FIG. **8**. In the present embodiment, the sealing and flow guiding component **124** has a plurality of positioning slots **124a**, and the bottle cap component **122** has two positioning ribs **1223** and two cutting edge ribs **1224**. The two positioning ribs **1223** respectively pass through a positioning slot **124a** and an air outlet hole **1242b**, and the two cutting edge ribs **1224** respectively pass through another two positioning slots **124a** to prevent the sealing and flow guiding component **124** from rotating relative to the bottle cap component **122**. Based on the above, in the present embodiment, the tire sealant holding bottle **50** and the bottle cap component **122** are engaged with each other in a relatively rotating manner, for example, by screwing the cap. When the tire sealant holding bottle **50** and the bottle cap component **122** are rotated relative to each other during the engaging process, the positioning design of the positioning slots **124a** and the positioning ribs **1223** may prevent the sealing and flow guiding component **124** from being twisted and deformed due to the rotation.

Moreover, during the engaging process of the tire sealant holding bottle **50** and the bottle cap component **122**, the cutting edge ribs **1224** are adapted to cut the sealing film on the bottleneck **52** of the tire sealant holding bottle **50** so that the sealing and flow guiding component **124** may smoothly communicate with the internal space of the tire sealant holding bottle **50**.

Please refer to FIG. **2** and FIG. **4**. In the present embodiment, the vehicle-mounted air compressor **100** further

includes a limiting component **140** and at least one fastening component **150** (two are shown). The limiting component **140** limits the bottle cap component **122** to the air compressor main body **110**. The fastening component **150** is, for example, a screw and fastens the limiting component **140** to the air compressor main body **110**. Accordingly, the user only needs to simply mount the bottle cap component **122** to the air compressor main body **110** and mount the limiting component **140** to complete the installation of the bottle cap component **122**, and the user only needs to simply remove the limiting component **140** and then remove the bottle cap component **122** to complete the disassembly of the bottle cap component **122** and/or the replacement of the tire sealant holding bottle **50**. More specifically, the fastening direction of the fastening component **150** is, for example, the same as the insertion direction (the direction D as shown in FIG. 2) of the air inlet end **1221** of the bottle cap component **122**, and the fastening position of the fastening component **150** and the insertion position of the air inlet end **1221** of the bottle cap component **122** are both located at the same side (that is, the upper side of the air compressor main body **110**) of the air compressor main body **110**, so that the user may readily assemble the vehicle-mounted air compressor **100** in the same position and in the same working direction. Moreover, since the fastening direction of the fastening component **150** is the same as the insertion direction of the air inlet end **1221** of the bottle cap component **122**, the air inlet end **1221** may be firmly inserted into the air compressor main body **110** by tightening the fastening component **150**.

The following describes the operation process of the vehicle-mounted air compressor of the present embodiment.

FIG. 12A to FIG. 12E are operation flowcharts of the vehicle-mounted air compressor of FIG. 1. FIG. 13A to FIG. 13C are top views of the vehicle-mounted air compressor of FIG. 12A to FIG. 12C respectively. First, as shown in FIG. 12A and FIG. 13A, the connecting pipe **130** is placed at the air outlet end **1222** of the bottle cap component **122**, and the connecting pipe **130** is fastened to the air outlet end **1222** using a pipe clamp **160**. Next, as shown in FIG. 12B and FIG. 13B, the bottle cap component **122** is mounted to the air compressor main body **110** via the guidance of two guiding surfaces **116** of the air compressor main body **110**, and the bottle cap component **122** is positioned by two hooks **114** of the air compressor main body **110**, and at the same time, the air inlet end **1221** of the bottle cap component **122** is inserted into the air outlet pipe **112** of the air compressor main body **110**, so that the air outlet pipe **112** is connected to the air inlet opening **122a**. The gap between the air inlet end **1221** and the air outlet pipe **112** may be sealed by a sealing ring **170** sleeved to the air inlet end **1221**. Moreover, the top surface of the air compressor main body **110** has an opening **110a**. As shown in FIG. 13B, the connection point between the air outlet pipe **112** and the bottle cap component **122** corresponds to the opening **110a**, allowing the user to visually confirm whether the air outlet pipe **112** and the bottle cap component **122** are truly connected via the opening **110a**.

Next, as shown in FIG. 12C and FIG. 13C, the limiting component **140** is fastened to the air compressor main body **110** via the fastening component **150** to achieve the state shown in FIG. 12D, so that the limiting component **140** presses the bottle cap component **122** against the air compressor main body **110** and the limiting component **140** presses the connecting pipe **130**. Next, as shown in FIG. 12D to FIG. 12E, the bottleneck **52** of the tire sealant holding bottle **50** is engaged with the bottle cap component **122**. During this process, a sealing film **52a** at the bottleneck **52**

is cut by the cutting edge ribs **1224** (shown in FIG. 7) of the bottle cap component **122**, so that the high-pressure airflow from the air compressor main body **110** may enter the tire sealant holding bottle **50** along the flow path P1 during a subsequent tire repair and inflation operation, and the tire sealant **501** in the tire sealant holding bottle **50** may leave the tire sealant holding bottle **50** along the flow path P2 with the high-pressure airflow during a subsequent tire repair and inflation operation. Lastly, as shown in FIG. 3, the connecting pipe **130** is connected to the tire **60**, and the tire **60** may be repaired and inflated via the vehicle-mounted air compressor **100**.

FIG. 14 shows that the tire sealant holding bottle in FIG. 1 is replaced by a cover body. FIG. 15 shows that the tire sealant holding bottle in FIG. 2 is replaced by a cover body. FIG. 16A shows that the tire sealant holding bottle in FIG. 12D is replaced by a cover body. FIG. 16B shows that the tire sealant holding bottle in FIG. 12E is replaced by a cover body. The vehicle-mounted air compressor **100** of the present embodiment may also only inflate the tire **60** without repairing the tire **60**. Specifically, the tire sealant holding bottle **50** in FIG. 1, FIG. 2, FIG. 12D, and FIG. 12E may be replaced by a cover body **70** in FIG. 14, FIG. 15, FIG. 16A, and FIG. 16B. The rest of the operation process thereof is the same as the above and is not repeated herein. Since the tire sealant holding bottle **50** is not used in the manner shown in FIG. 14, FIG. 15, 16A, and FIG. 16B, after passing through the bottle cap assembly **120**, the high-pressure airflow from the air compressor main body **110** directly reaches the tire for inflation, as shown in a flow path P3 of FIG. 16B.

Based on the above, in the vehicle-mounted air compressor of the invention, in addition to providing the existing sealing function via the sealing portion thereof, the sealing and flow guiding component also covers the air inlet opening of the bottle cap component via the flow guiding portion thereof. Accordingly, while the high-pressure airflow from the air compressor main body flows smoothly to the air outlet opening of the bottle cap component via the guidance of the flow guiding portion, the flow guiding portion may prevent the tire sealant in the tire sealant holding bottle from flowing to the air inlet opening of the bottle cap component. That is, in the invention, the flow guiding portion is integrally formed on the existing sealing component to cover the air inlet opening of the bottle cap component, so no additional parts are needed to achieve this object. Therefore, the vehicle-mounted air compressor and the bottle cap assembly thereof of the invention prevent the tire sealant from flowing to the air compressor main body via a simple structure, thereby reducing the difficulty of manufacturing and assembling the bottle cap assembly.

What is claimed is:

1. A vehicle-mounted air compressor, comprising:
 - an air compressor main body; and
 - a bottle cap assembly, comprising:

- a bottle cap component mounted at the air compressor main body and having an air inlet opening and an air outlet opening, wherein the bottle cap component is connected to the air compressor main body via the air inlet opening; and

- a sealing and flow guiding component disposed at the bottle cap component and comprising a sealing portion and a flow guiding portion, wherein the sealing portion surrounds the flow guiding portion, the flow guiding portion covers the air inlet opening and has at least one air inlet hole and an air outlet hole, the flow guiding portion is connected to the air inlet opening via the at least one air inlet hole and

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connected to the air outlet opening via the air outlet hole, and an airflow from the air compressor main body is adapted to pass through the air inlet opening, the at least one air inlet hole, the air outlet hole, and the air outlet opening in sequence,

wherein a flow guiding channel is formed between the sealing and flow guiding component and the bottle cap component, and the flow guiding channel is connected between the at least one air inlet hole and the air outlet hole.

2. The vehicle-mounted air compressor of claim 1, wherein the sealing portion and the flow guiding portion are integrally formed structures.

3. The vehicle-mounted air compressor of claim 1, wherein a width of the flow guiding channel is less than an inner diameter of the air inlet opening and an inner diameter of the at least one air inlet hole.

4. The vehicle-mounted air compressor of claim 1, wherein the flow guiding channel is extended in a zigzag manner.

5. The vehicle-mounted air compressor of claim 1, wherein the flow guiding portion comprises a boss, the at least one air inlet hole is located at a top end of the boss, and a bottom end of the boss is adjacent to the air outlet hole and the sealing portion.

6. The vehicle-mounted air compressor of claim 1, wherein the sealing and flow guiding component has at least one positioning slot, the bottle cap component has at least one positioning rib, and the at least one positioning rib passes through the at least one positioning slot to prevent the sealing and flow guiding component from rotating relative to the bottle cap component.

7. The vehicle-mounted air compressor of claim 1, wherein the bottle cap component is adapted to engage with a bottleneck of a tire sealant holding bottle, so that the sealing portion is compressed between the bottle cap component and the bottleneck of the tire sealant holding bottle, and the air inlet hole and the air outlet hole are connected to an internal space of the tire sealant holding bottle.

8. The vehicle-mounted air compressor of claim 7, wherein the bottle cap component has at least one cutting edge rib, and the at least one cutting edge rib is adapted to cut a sealing film on the bottleneck of the tire sealant holding bottle.

9. The vehicle-mounted air compressor of claim 1, further comprising a limiting component and at least one fastening component, wherein the limiting component limits the bottle cap component to the air compressor main body, and the at least one fastening component fastens the limiting component to the air compressor main body.

10. The vehicle-mounted air compressor of claim 1, wherein the air compressor main body has an air outlet pipe, the air outlet pipe is connected to the bottle cap component to be connected to the air inlet opening, there is an opening at a top surface of the air compressor main body, and a connection point between the air outlet pipe and the bottle cap component corresponds to the opening.

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11. A bottle cap assembly, adapted for a vehicle-mounted air compressor, comprising:

a bottle cap component mounted at an air compressor main body of the vehicle-mounted air compressor and having an air inlet opening and an air outlet opening, wherein the bottle cap component is connected to the air compressor main body via the air inlet opening; and

a sealing and flow guiding component disposed at the bottle cap component and comprising a sealing portion and a flow guiding portion, wherein the sealing portion surrounds the flow guiding portion, the flow guiding portion covers the air inlet opening and has at least one air inlet hole and an air outlet hole, the flow guiding portion is connected to the air inlet opening via the at least one air inlet hole and connected to the air outlet opening via the air outlet hole, and an airflow from the air compressor main body is adapted to pass through the air inlet opening, the at least one air inlet hole, the air outlet hole, and the air outlet opening in sequence, wherein a flow guiding channel is formed between the sealing and flow guiding component and the bottle cap component, and the flow guiding channel is connected between the at least one air inlet hole and the air outlet hole.

12. The bottle cap assembly of claim 11, wherein the sealing portion and the flow guiding portion are integrally formed structures.

13. The bottle cap assembly of claim 11, wherein a width of the flow guiding channel is less than an inner diameter of the air inlet opening and an inner diameter of the at least one air inlet hole.

14. The bottle cap assembly of claim 11, wherein the flow guiding channel is extended in a zigzag manner.

15. The bottle cap assembly of claim 11, wherein the flow guiding portion comprises a boss, the at least one air inlet hole is located at a top end of the boss, and a bottom end of the boss is adjacent to the air outlet hole and the sealing portion.

16. The bottle cap assembly of claim 11, wherein the sealing and flow guiding component has at least one positioning slot, the bottle cap component has at least one positioning rib, and the at least one positioning rib passes through the at least one positioning slot to prevent the sealing and flow guiding component from rotating relative to the bottle cap component.

17. The bottle cap assembly of claim 11, wherein the bottle cap component is adapted to engage with a bottleneck of a tire sealant holding bottle, so that the sealing portion is compressed between the bottle cap component and the bottleneck of the tire sealant holding bottle, and the air inlet hole and the air outlet hole are connected to an internal space of the tire sealant holding bottle.

18. The bottle cap assembly of claim 17, wherein the bottle cap component has at least one cutting edge rib, and the at least one cutting edge rib is adapted to cut a sealing film on the bottleneck of the tire sealant holding bottle.

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